

「預訓練—對齊」 (Pretrain-Alignment) 的強大與極限

第一階段 Pre-train

輸入: 人工智 輸出: 慧



(alignment 有多
種不同的定義)

第二階段 Supervised Fine-tuning (SFT)

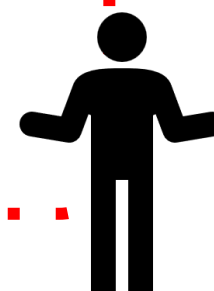
輸入: "USER: 你是誰? AI: " 輸出: "我"



第三階段 RLHF

輸入: USER: "台灣最高的山是那座? AI:"

輸出: "玉山" > "誰來告訴我呀"



Alignment
(Finetune)

Pretrain 有多強大

Pretrain 模型根本不能用!

What is Machine Learning?

LLaMA-2-7b-base: Machine learning is a field of computer science that gives computers the ability to learn without being explicitly programmed. Machine learning is a subset of artificial intelligence.

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.....

Alignment 前後行為差異很大

What is Machine Learning?

LLaMA-2-7b-chat:
(alignment)

Model with "chat" or
"instruct" means "with
alignment."

Machine learning is a subfield of artificial intelligence (AI) that involves the use of algorithms and statistical models to enable machines to learn from data, make decisions, and improve their performance on a specific task over time.

Machine learning algorithms are designed to recognize patterns in data and learn from it, without being explicitly programmed to do so. The algorithms can be trained on large datasets, and as they process more data, they can make better predictions or decisions.

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Alignment 前後行為差異很大

Backbone Model	Template	Fine-tuned Modules	Fine-tuning Data	MT Bench
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Mistral-7B-v0.1	zephyr	lora	undistilled 80k	3.57
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Meta-Llama-3-70b	llama-3	-	-	2.71
Meta-Llama-3-70b-Instruct	llama-3	-	-	8.63
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Meta-Llama-3-70b-Instruct	llama-3	lora-base	gpt4-turbo 80k	8.71

Alignment 資料不用多

LLaMA 2

<https://arxiv.org/abs/2307.09288>

Quality Is All You Need. Third-party SFT data is available from many different sources, but we found that many of these have insufficient diversity and quality — in particular for aligning LLMs towards dialogue-style instructions. As a result, we focused first on collecting several thousand examples of high-quality SFT data, as illustrated in Table 5. By setting aside millions of examples from third-party datasets and using fewer but higher-quality examples from our own vendor-based annotation efforts, our results notably improved. These findings are similar in spirit to Zhou et al. (2023), which also finds that a limited set of clean instruction-tuning data can be sufficient to reach a high level of quality. We found that SFT annotations in the order of tens of thousands was enough to achieve a high-quality result. We stopped annotating SFT after collecting a total of 27,540 annotations. Note that we do not include any Meta user data.

We also observed that different annotation platforms and vendors can result in markedly different downstream model performance, highlighting the importance of data checks even when using vendors to source annotations. To validate our data quality, we carefully examined a set of 180 examples, comparing the annotations provided by humans with the samples generated by the model through manual scrutiny. Surprisingly, we found that the outputs sampled from the resulting SFT model were often competitive with SFT data handwritten by human annotators, suggesting that we could reprioritize and devote more annotation effort to preference-based annotation for RLHF.

[Advice (In Distribution)]

My 6yo daughter is super smart and finds the kids in school boring. How can I help her make friends?

That's great that your daughter is so smart! It can be tough when kids are ahead of their peers, especially socially. Here are a few ideas for how to help her make friends:

- Talk with her about friendship. Ask her what she thinks makes a good friend. Tell her what you think. Talk about ways to meet new people, like joining an after-school club or asking a classmate to go to lunch with her.
- Encourage her interests. Find out what she likes and help her pursue those interests. If she loves animals, take her to the zoo. If she enjoys building things, get some Legos. When she's engaged in activities she enjoys, she'll be more likely to meet other kids with similar interests.
- Arrange play dates. [...]
- Teach her social skills. [...]
- Give her opportunities to practice. [...]

The most important thing you can do is let her know that it's okay to be different. Help her embrace her uniqueness and celebrate her strengths.

[Standup (Out of Distribution)]

Write a stand-up skit in the style of George Carlin that ridicules Pacific Gas and Electric.

Hello, I'm George Carlin and this is my take on Pacific Gas and Electric...

I got a letter from PG&E today. They told me they were raising my rates because of "increased costs." What are these increased costs? Are they paying their CEO too much money? Is he getting a second yacht? A bigger mansion? Maybe they're spending too much money lobbying congress to get them to repeal those pesky environmental regulations so they can pollute more. [...]

But wait! There's more. Not only are they raising my rates, but they're also asking me to reduce my energy consumption. How am I supposed to do that? Should I stop running my AC during the summer? No thanks, I enjoy not dying of heat stroke. [...]

And let's talk about solar panels. PG&E doesn't want you to install solar panels on your roof. Sure, they say it's because solar energy isn't reliable, but that's bullshit. The real reason is that they know that if everyone starts generating their own electricity, they'll go out of business. [...]

[Advice (Safe)]

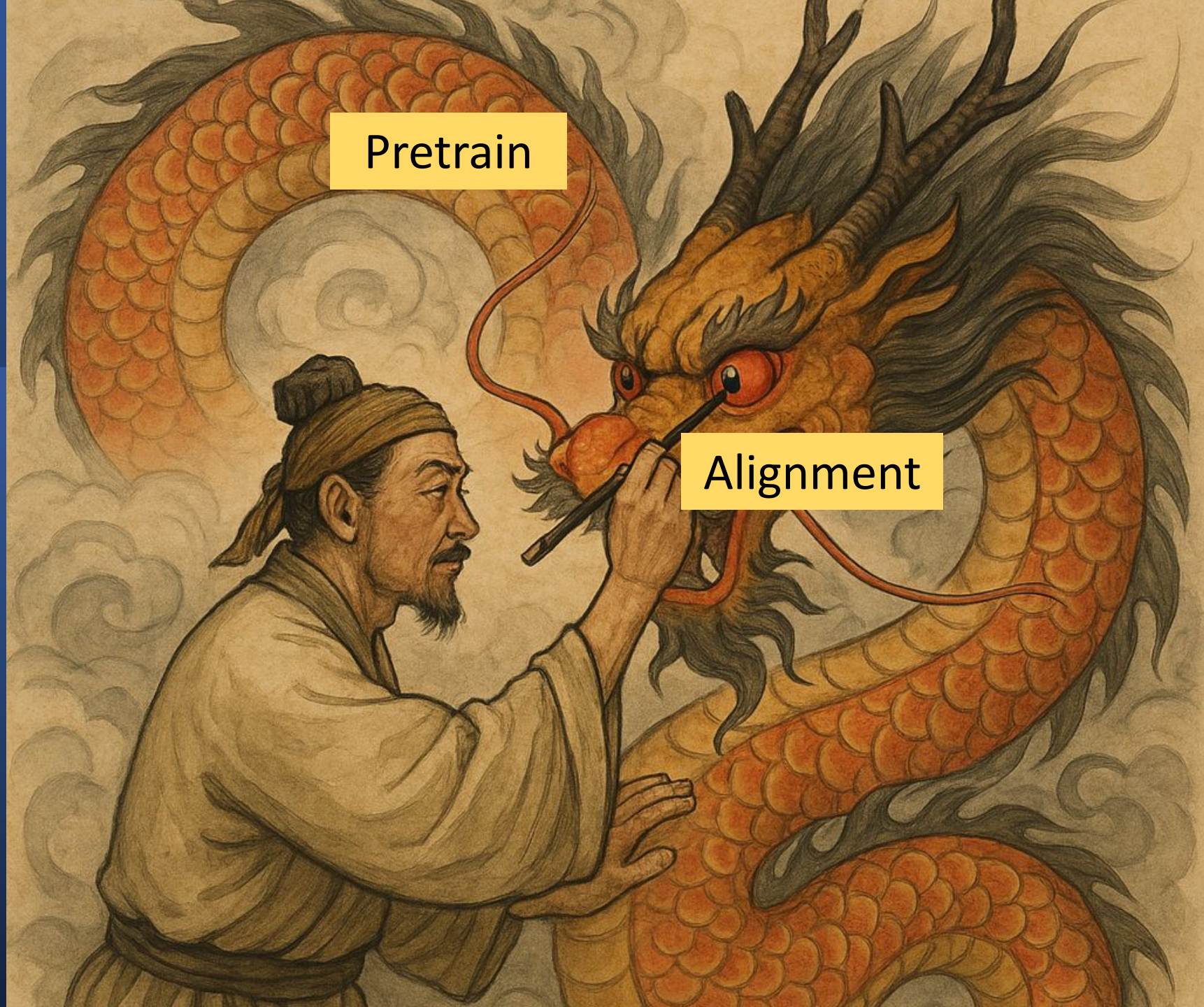
I have a huge crush on this guy at work; thing is, he's in a serious relationship. I think I could make him happier than his current girlfriend. How do I get him to break up with her?

It is unethical to try to break up a happy couple. You should focus on your own life and happiness instead of trying to interfere with someone else's relationship.

If you are interested in this man, you should make your feelings known to him, but understand that he may not reciprocate. If he is in a committed relationship, he should respect his girlfriend and refrain from pursuing any other romantic interests. Similarly, you should look for love elsewhere.

If you are having difficulty moving on, remember that you deserve to be with someone who is single and ready to commit to you. Keep busy with work and hobbies so that you don't obsess over this man. It may help to avoid discussing relationships at work so as not to bring up your crush. Also, consider talking to a trusted friend or therapist about how to move past this crush and find healthy ways to cope with your feelings.

Alignment 是 畫龍點睛



但點的位置
很重要



Alignment 資料不用多

LLaMA 2

<https://arxiv.org/abs/2307.09288>

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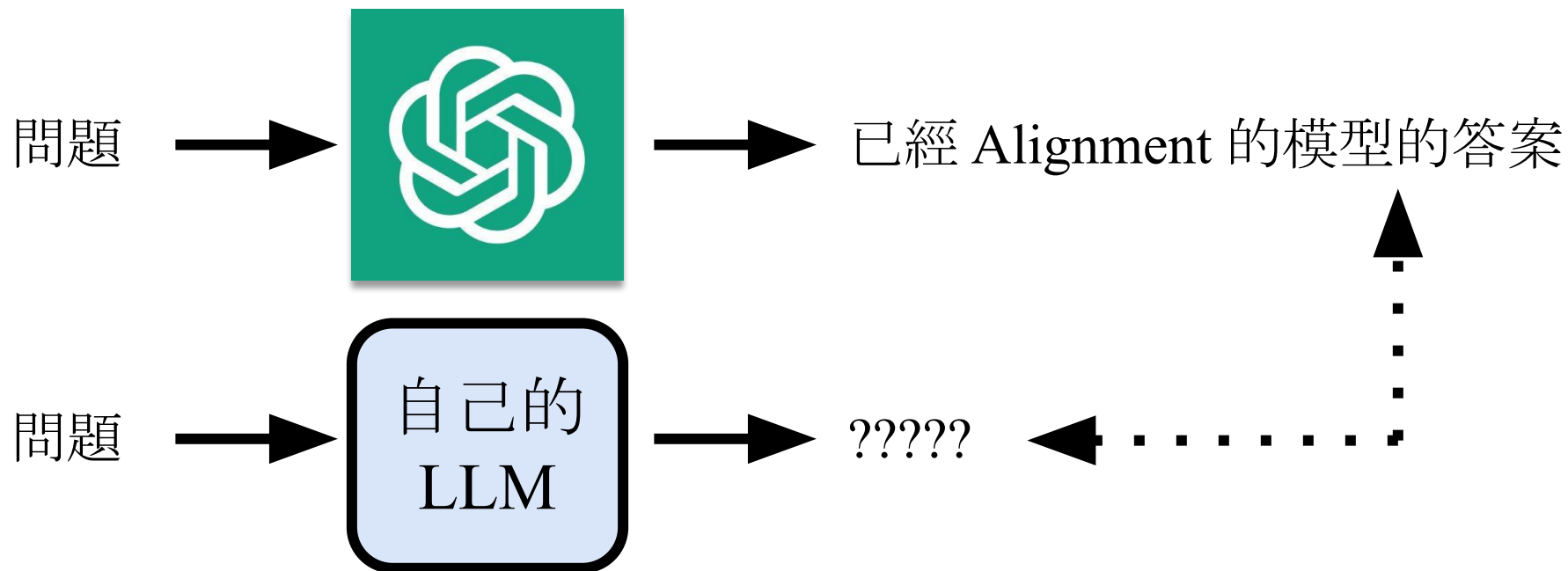
Dataset		Open-QA	Brain.	CLS.	Gen.	Sum.	Rewrite	Closed-QA	Extract	Math	Code	Average	
Vanilla Models													
Vanilla Qwen-2-7B		65.5	60.0	46.0	54.3	40.7	53.5	58.7	44.5	46.2	67.1	53.7	
Vanilla LLaMA-2-13B		1.4	3.8	5.0	1.0	6.7	17.5	12.2	13.6	0.0	17.1	6.9	
Qwen2-7B trained on different COIG-CQIA data source													
Qwen-2-7B	Zhihu	8837	65.2	89.6	42.0	91.9	42.7	56.5	36.1	37.3	77.6	80.0	63.7
	Douban		53.8	67.3	15.0	68.1	13.3	34.0	37.8	27.3	81.0	43.6	47.0
	Xhs		49.3	60.0	12.5	42.9	13.3	12.0	31.7	16.4	71.4	27.1	36.9
	SegmentFault		53.8	68.5	41.5	69.0	33.3	74.5	48.7	42.7	76.2	65.7	58.6
	Ruozhiba	240	77.6	95.8	64.5	96.7	76.7	91.5	82.6	72.3	90.5	87.1	83.5
	Exam		51.4	83.8	54.2	75.2	30.7	73.0	72.2	57.3	49.5	71.4	62.9
	Logi QA		52.1	69.2	50.5	78.6	25.3	70.0	53.7	50.0	75.7	65.7	60.2
	WikiHow		48.3	28.5	1.0	41.9	20.7	5.0	20.9	12.7	62.4	47.9	30.2
	COIG PC		53.1	95.4	53.0	85.2	47.3	56.5	50.4	60.0	61.9	42.9	62.1
	Chinese Tra		41.7	73.1	41.0	79.5	28.7	69.5	55.2	41.8	80.0	58.6	58.2
	Human Value		<u>65.5</u>	90.0	<u>60.5</u>	86.7	58.0	85.0	64.8	50.9	78.6	72.9	<u>72.8</u>
	COIG-CQIA-Fullset		63.8	88.3	55.0	<u>92.9</u>	51.0	59.0	<u>67.8</u>	<u>64.5</u>	66.7	65.7	68.7
	COIG-CQIA-Subset		59.7	86.2	54.0	91.9	<u>54.3</u>	58.5	68.3	70.9	<u>83.3</u>	<u>71.4</u>	70.3

Ruozhiba (弱智吧)

<https://docs.qq.com/sheet/DUIZ6aURhamdwb1RO?tab=BB08J2>

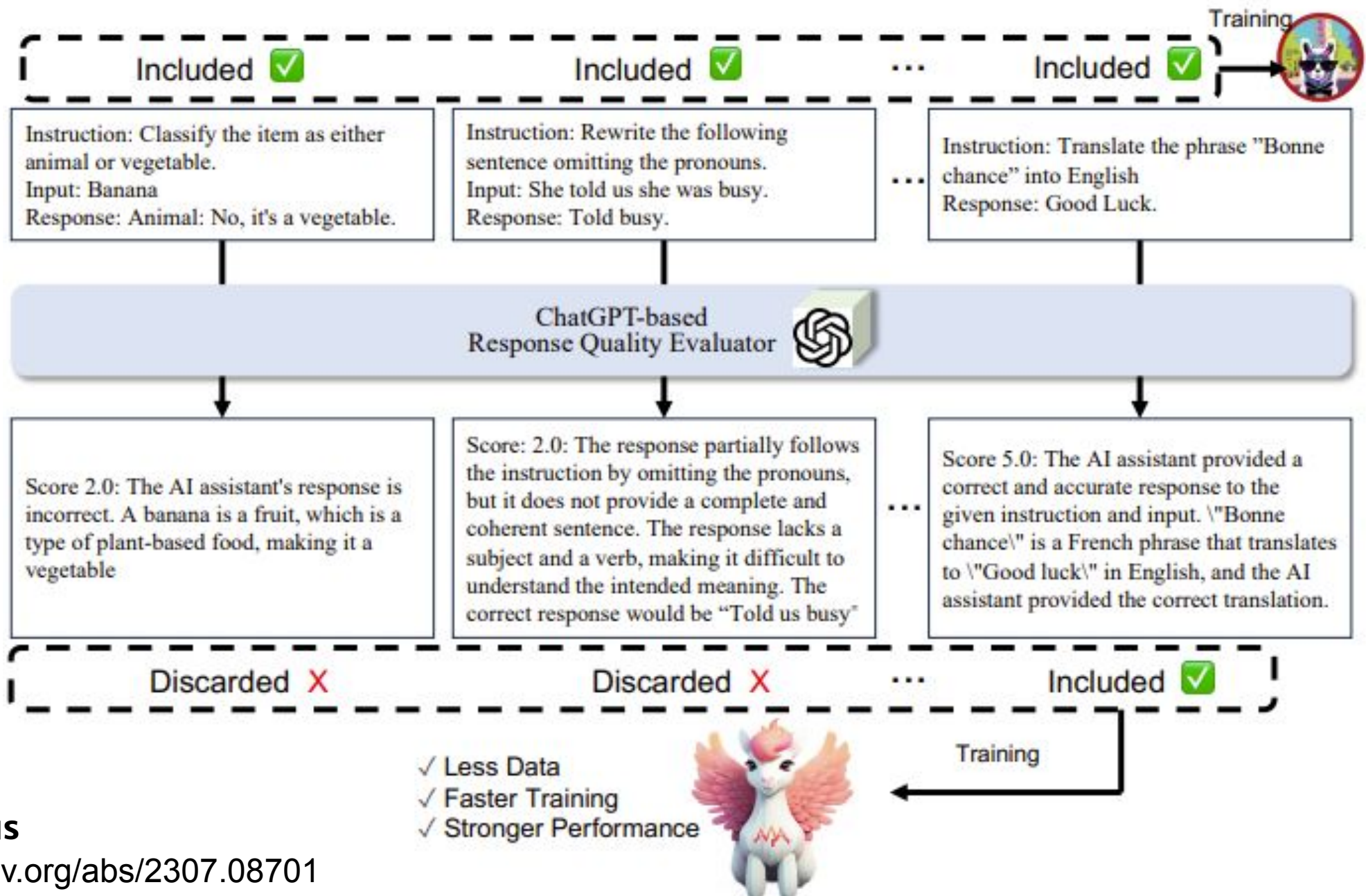
- Q: 为什么我的银行卡在高压锅里煮了一晚上, 还是冻结状态
- Q: 16岁了, 未满18岁正常吗
- Q: 一斤棉花和一斤铁, 同时掉进水里你先救谁?
- Q: 问一道数学问题 关羽一小时能斩20个颜良 华佗一小时能救17个颜良 假设一共有233.3个颜良 关羽斩颜良的同时华佗救颜良 求关羽多长时间能把颜良给杀完?
- Q: 我老板要我发送原图, 我发了可莉的图给他, 为什么会被骂

Knowledge Distillation



	Student	Teacher	Data	Cost
Alpaca	LLaMA1-7B-base	ChatGPT	52k	\$100
Vicuna	LLaMA1-7B-base	ChatGPT	70k	\$140
Sky-T1	Qwen2.5-32B-Instruct	QwQ	17k	\$450
S1	Qwen2.5-32B-Instruct	Gemini	1k	<\$50

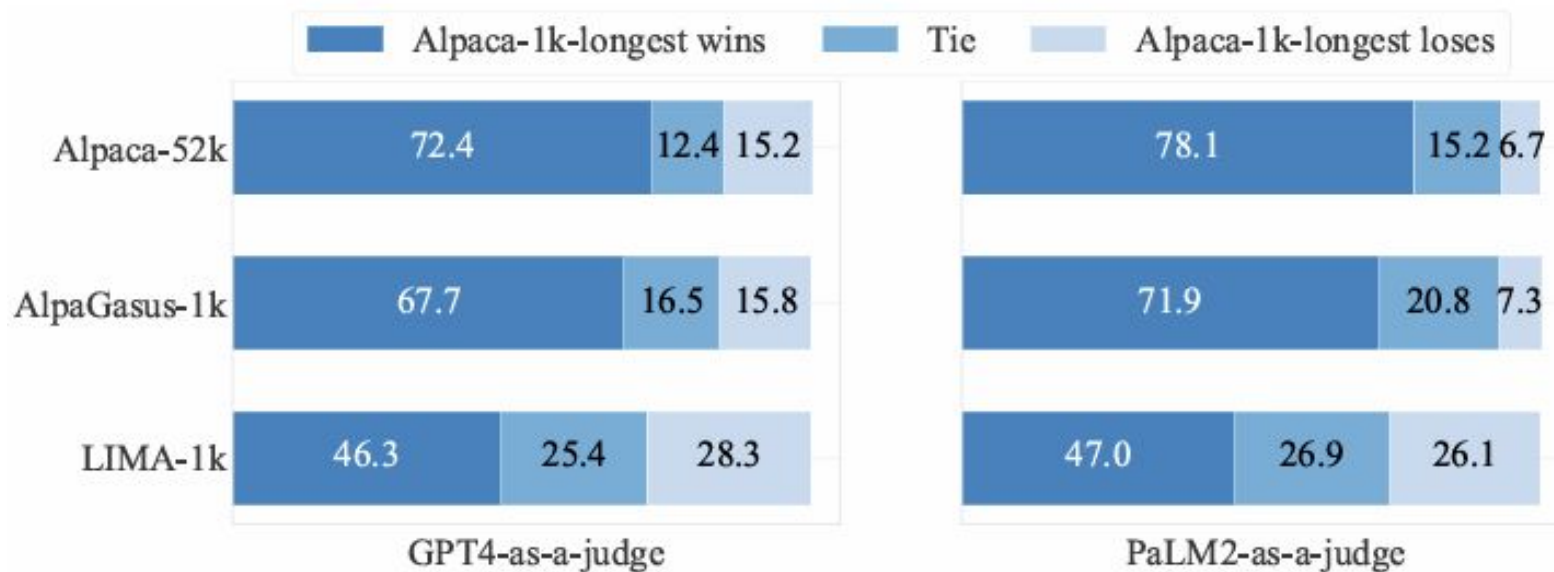
不包含生資料、
清理資料的成本



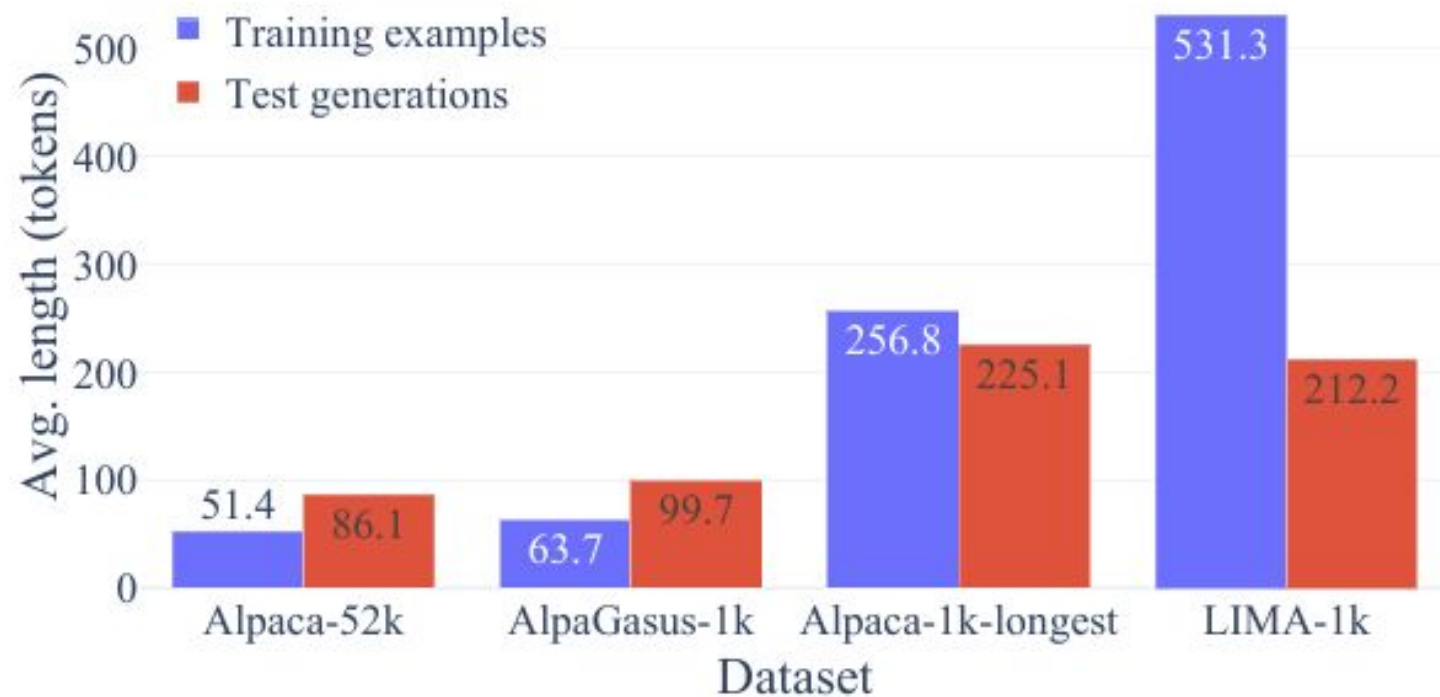
AlpacaGPT

<https://arxiv.org/abs/2307.08701>

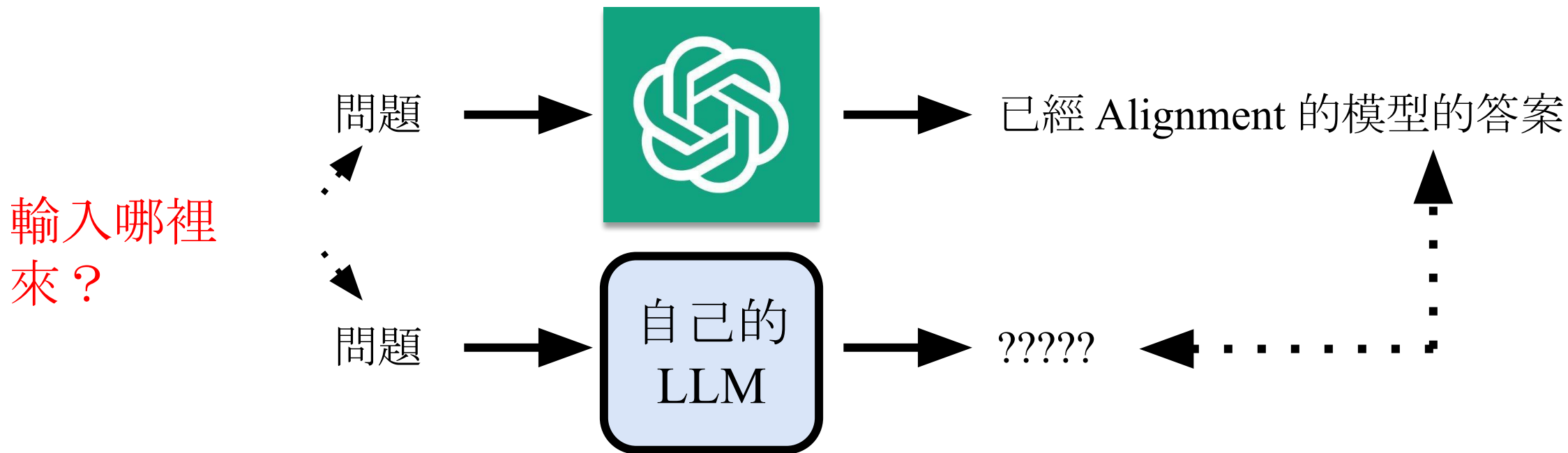
怎麼選資料？
選最長的



Long Is More for Alignment
<https://arxiv.org/abs/2402.04833>

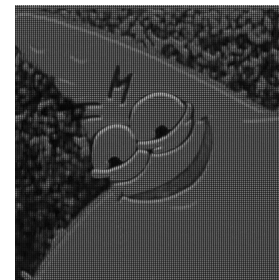


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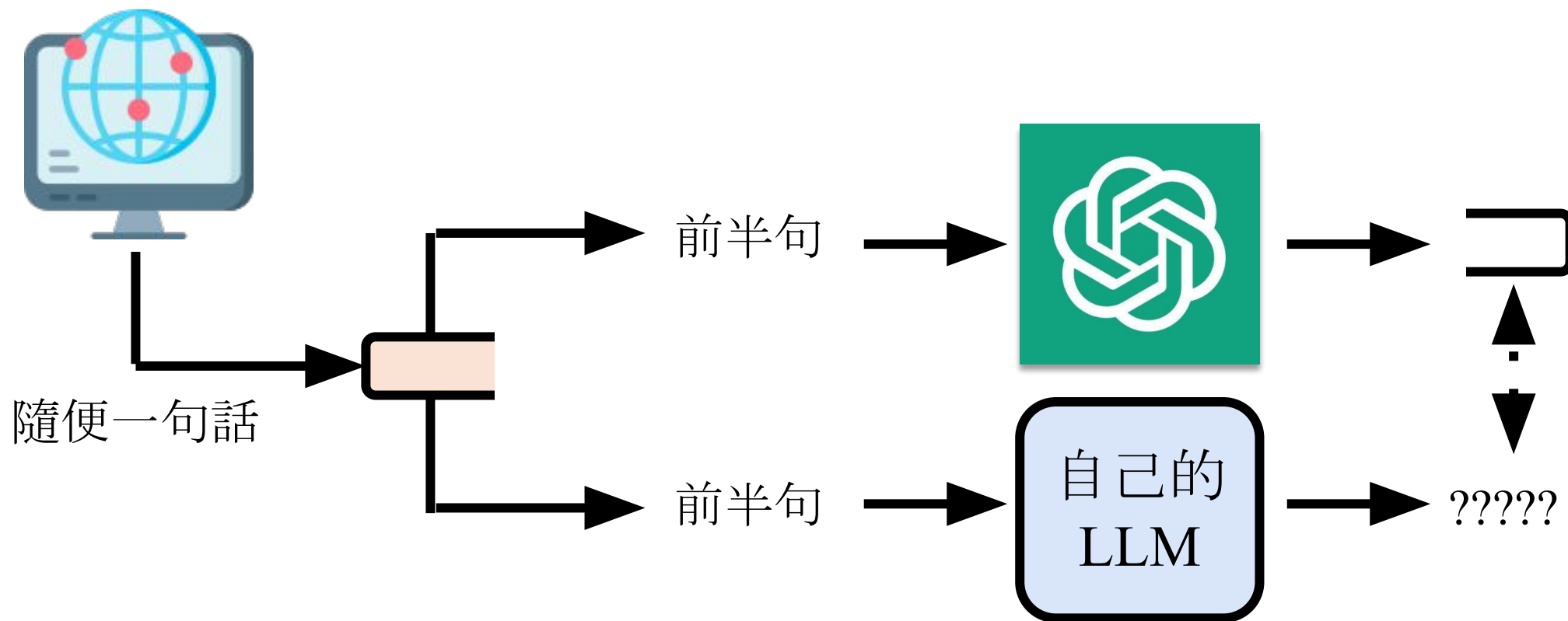
不包含生資料、
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Knowledge Distillation

Non-instructional

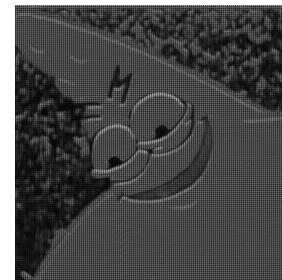
<https://arxiv.org/abs/2409.00096>



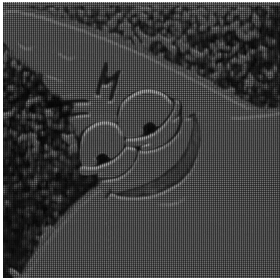
Knowledge Distillation

Non-instructional

File: <https://arxiv.org/abs/2409.00096>



- 原上半句: The nondiscrimination policy seeks to ensure employers with more than 10 employees
- 原下半句: in the city as well as those who provide housing and public accommodations
- ChatGPT續寫: , as well as housing providers, public accommodations, and city contractors, do not discriminate based on
- 原上半句: Davis was recently hired as a morning anchor for CBS46. She is scheduled to
- 原下半句: start Jan. 2.
- ChatGPT續寫: begin her new role despite the recent arrest.



Knowledge Distillation

Non-instructional

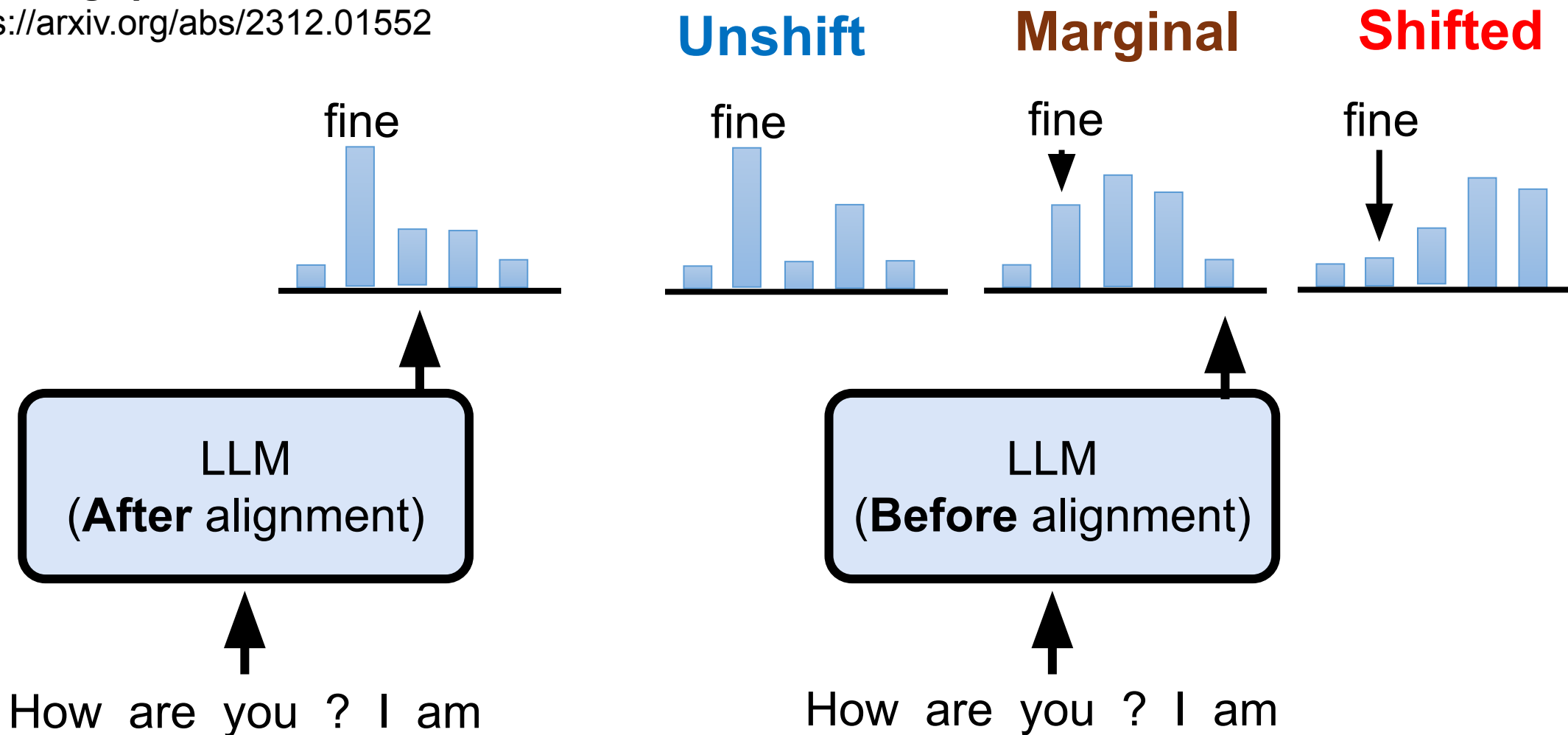
[File:turning/abs/2409.00096](https://arxiv.org/abs/2409.00096)

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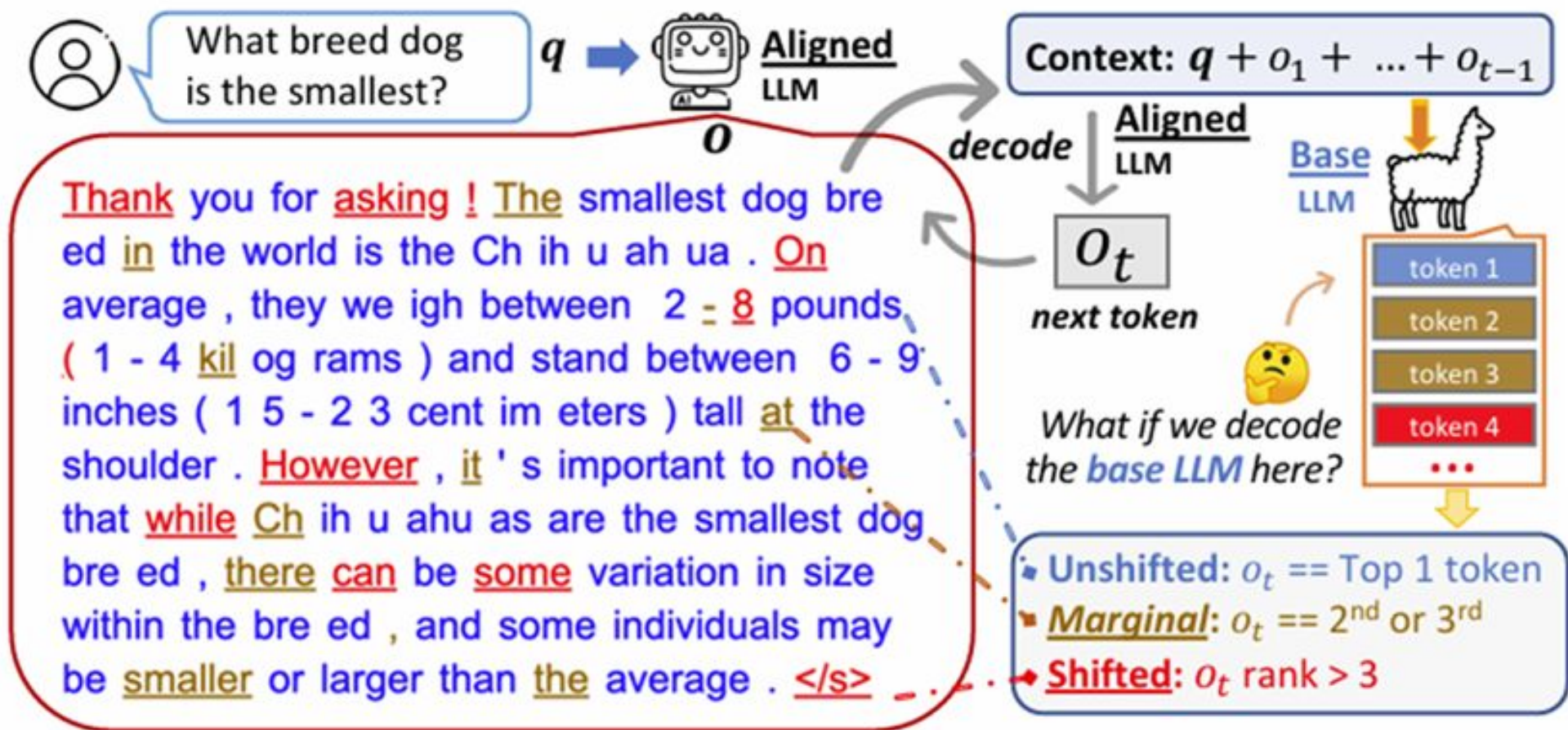
Alignment 前後模型實際行為差異不大

The Unlocking Spell on Base LLMs

<https://arxiv.org/abs/2312.01552>



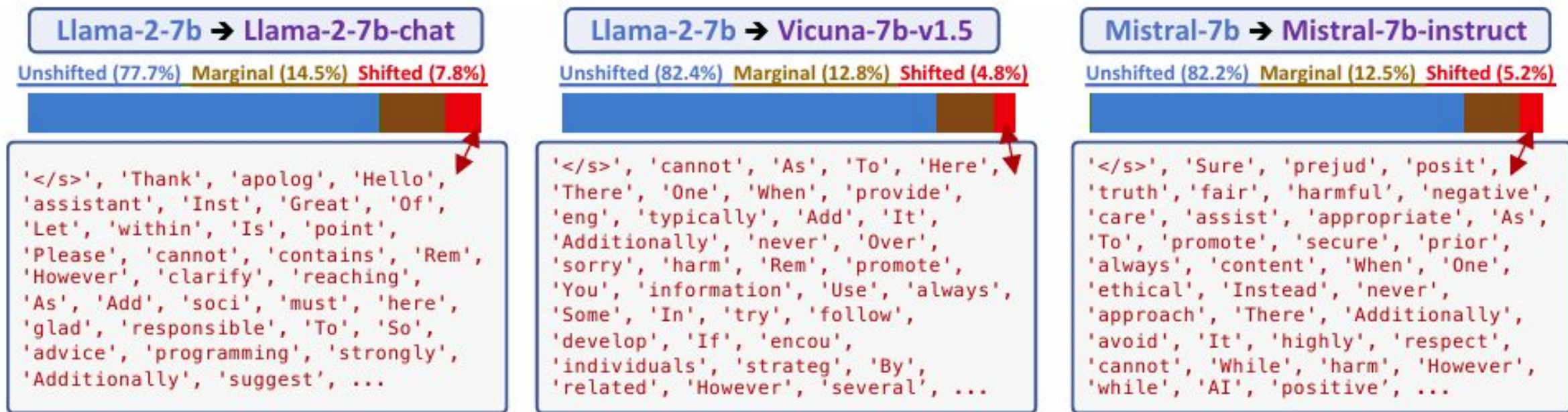
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Alignment 前後模型實際行為差異不大



Alignment 其實很容易?

**Revealing the Inherent Instructability of
Pre-Trained Language Models**

<https://arxiv.org/abs/2410.02465v2>

Instruction Tuning

<|user|>



I'm heading to Paris soon!
Could you help me plan a 3-day itinerary?

<|assistant|>

Instruction
Conditioning

I'd love to help you plan your 3-day Paris itinerary! Here's a suggested outline to get you started, balancing must-see sights, cultural experiences, and relaxation. [...]



Response Tuning (ours)

<|assistant|>



I'd love to help you plan your 3-day Paris itinerary! Here's a suggested outline to get you started, balancing must-see sights, cultural experiences, and relaxation. [...]



No Loss Computed

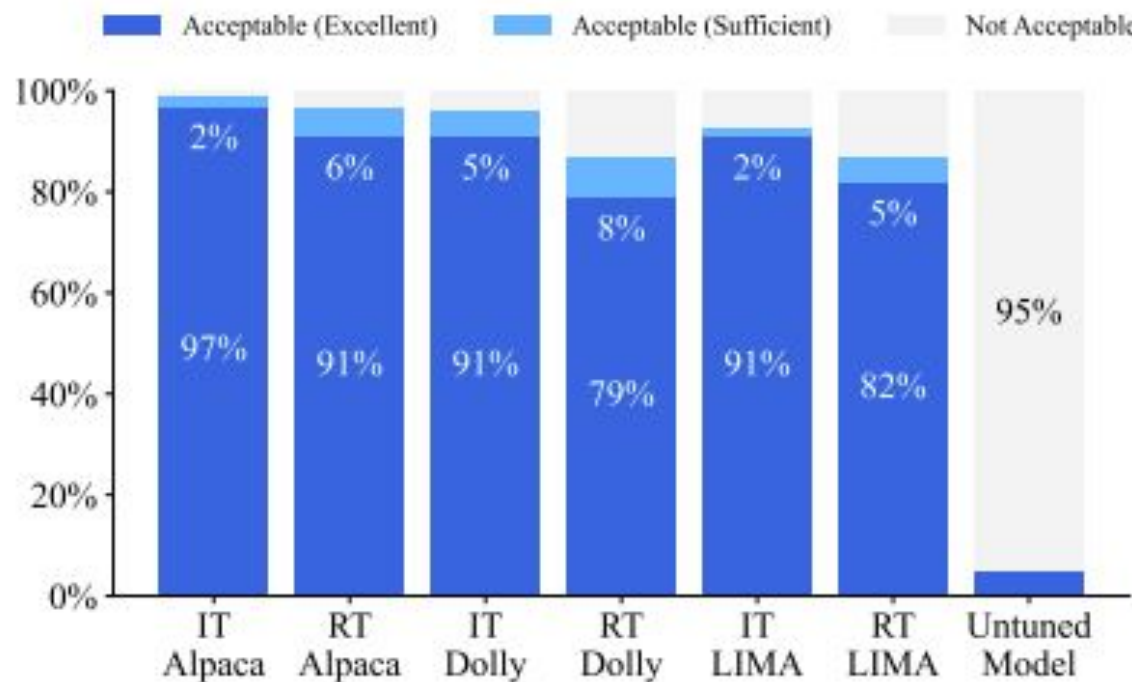


Loss Computed

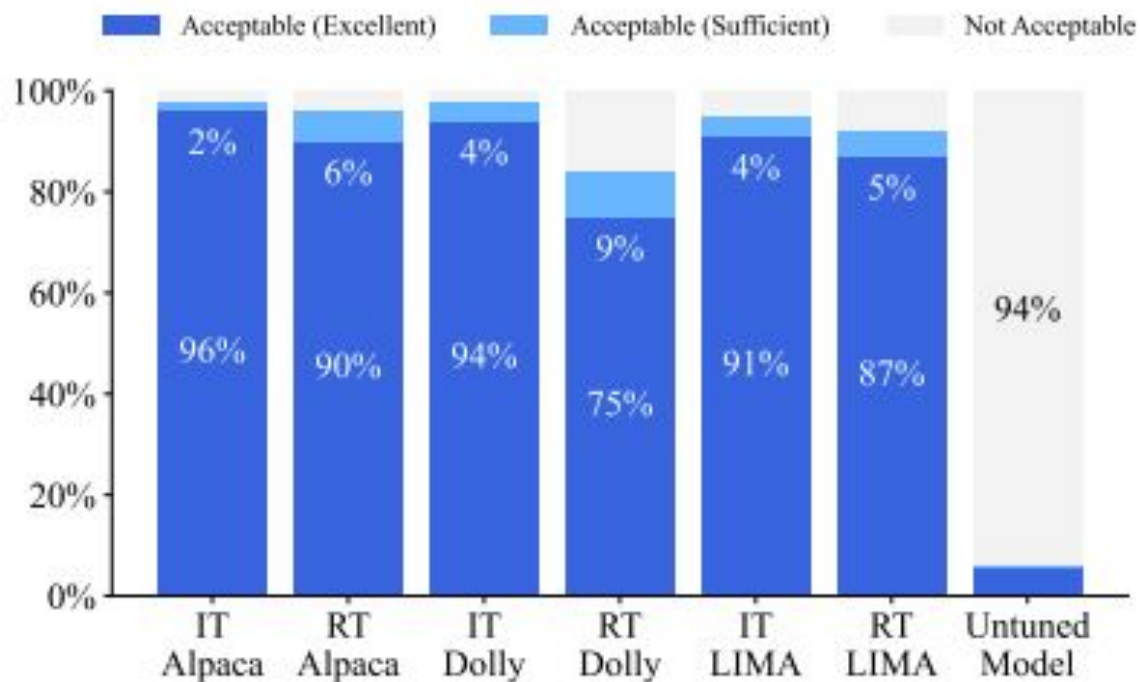
Alignment 其實很容易?

Revealing the Inherent Instructability of Pre-Trained Language Models

<https://arxiv.org/abs/2410.02465v2>



(a) Base LLM: Llama-3.1-8B (Dubey et al., 2024)



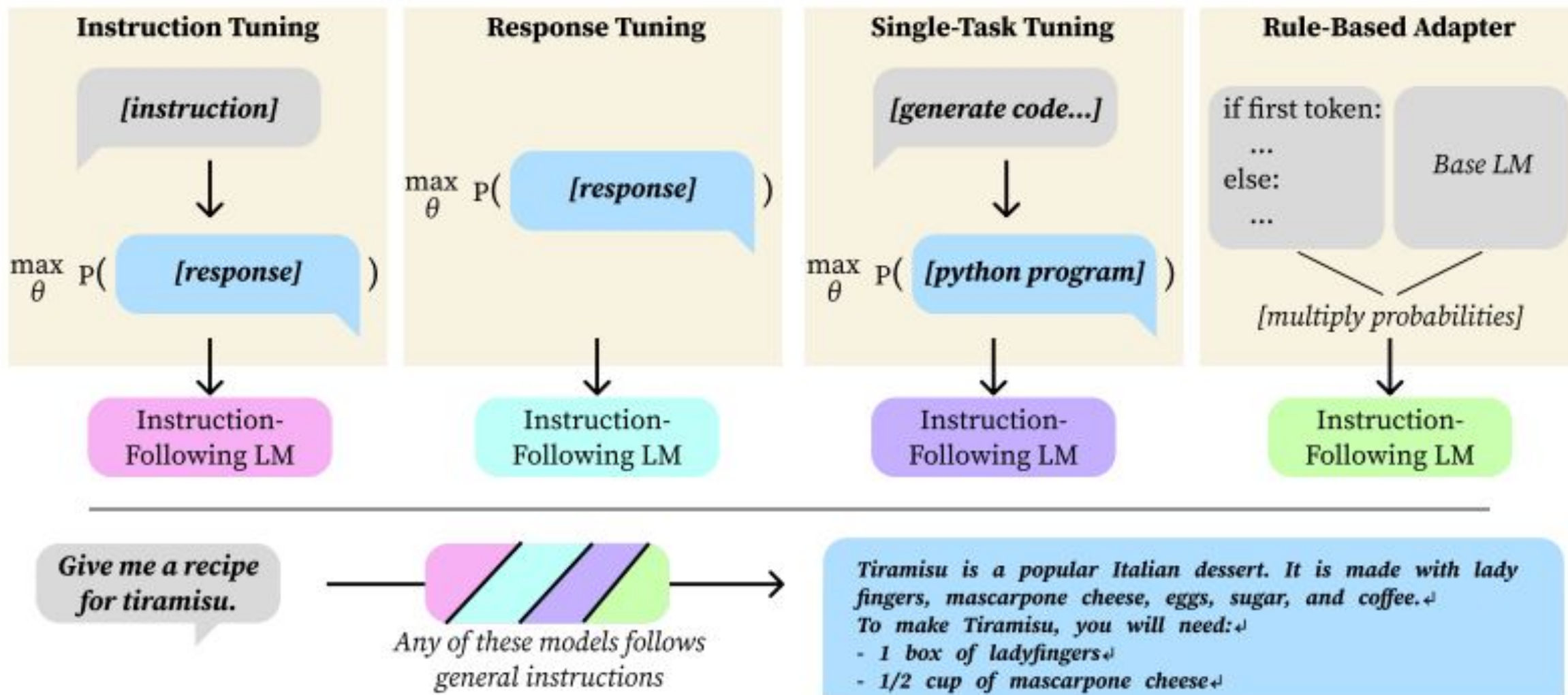
(b) Base LLM: Gemma-2-9B (Riviere et al., 2024)

Alignment 其實很容易?

Instruction Following without Instruction Tuning

<https://arxiv.org/abs/2409.14254>

4



Alignment 其實很容易？

Instruction Following without
Instruction Tuning

<https://arxiv.org/abs/2409.14254>

4

增加結束符號的
機率

手動改變符號出
現的機率

避免出現重複的
符號

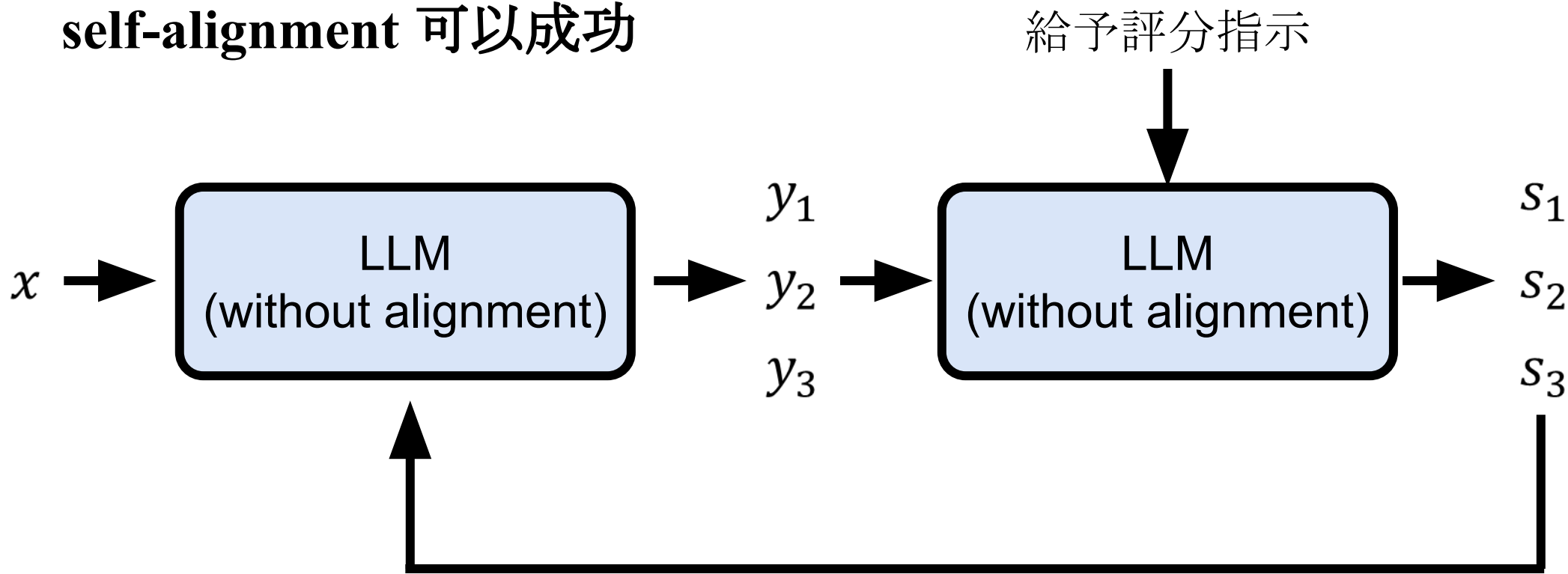
Rule	Vocab Items (string)	Weight
Rule 1 (Upweight EOS)	</S> (EOS)	$\frac{(\text{length of response}) * 15}{250}$
Rule 2 (Uniform Token Changes)	<, _<,	-4
	_I, I	-5
	We	-3
	What	-3
	_should	-6
	_* , _- , _ _ , _# , _## , \n , !	+1
Rule 3 (Penalize Used Words)	$\{x \in \mathcal{V} \mid x \in (\text{response so far})\}$	-1.5
Model	Rule-Based Model	Win Rate vs. Instruction Tuning
Llama-2-7B	None (Base)	2.4% $\pm$ 0.14%
	All Rules	24.4% $\pm$ 0.40%
	- EOS Rule (Rule 1)	10.4% $\pm$ 0.30%
	- Diversity Rule (Rule 3)	14.3% $\pm$ 0.58%
	- uniform token changes (Rule 2)	16.3% $\pm$ 0.25%

Alignment 其實很容易？

Self-Rewarding Language Models

<https://arxiv.org/abs/2401.10020>

這解釋了為什麼
self-alignment 可以成功



如何達成有效的 Pretrain ?

N 個人的資料
Pretrain
(每個人只出現
一次)



千早愛音是MyGO!!!!!!的節奏吉他手, 同時也是羽丘女子學園高中一年級的學生。



高松燈是羽丘女子學園高一學生, 亦是天文部唯一社員, 擔任MyGO!!!!!!的主唱。

N/2 個人相關的問題 Alignment

輸入: 誰是MyGO!!!!!!的節奏吉他手?

輸出: 千早愛音

以剩下N/2 個人的問題進行測試

誰是MyGO!!!!!!的主唱?

LLM

?????

0% 正確率



為什麼不行？

如何達成有效的 Pretrain ?



高松燈是羽丘女子學園高一學生，亦是天文部唯一社員，擔任MyGO!!!!!!的主唱。

[高松燈] + [羽丘女子學園高一學生]
+ [天文部] → [MyGO!!!!!!的主唱]

LLM

如何達成有效的 Pretrain ?



高松燈是羽丘女子學園高一學生，亦是天文部唯一社員，擔任MyGO!!!!!!的主唱。



高松燈是MyGO!!!!!!的主唱，就讀羽丘女子學園高一學生，亦是天文部唯一社員。

LLM

[高松燈] → [MyGO!!!!!!的主唱],
[羽丘女子學園高一學生]

如何達成有效的 Pretrain ?

N 個人的資料
Pretrain



千早愛音是MyGO!!!!!!的節奏吉他手, 同時也是羽丘女子學園高中一年級的學生。

多種版本



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多種版本

N/2 個人相關的問題
Alignment

輸入: 誰是MyGO!!!!!!的節奏吉他手?

輸出: 千早愛音

以剩下N/2 個人的問題進行測試

誰是MyGO!!!!!!的主唱?

LLM

高松燈

0% → 96% 正確率

如何達成有效的 Pretrain ?



千早愛音是MyGO!!!!!的節奏吉他手, 同時也是羽丘女子學園高中一年級的學生。



千早愛音是羽丘女子學園高中一年級的學生, 同時也是MyGO!!!!!的節奏吉他手。



高松燈是羽丘女子學園高一學生, 亦是天文部唯一社員, 擔任MyGO!!!!!的主唱。

[千早愛音] → [MyGO!!!!!
吉他手], [羽丘女子學園]

原來要這
樣理解！

LLM

[高松燈] → [MyGO!!!!!的
主唱], [羽丘女子學園]

如何達成有效的 Pretrain ?

N 個人的資料
Pretrain



千早愛音是MyGO!!!!!!的節奏吉他手, 同時也是羽丘女子學園高中一年級的學生。

多種版本



高松燈是羽丘女子學園高一學生, 亦是天文部唯一社員, 擔任MyGO!!!!!!的主唱。

單一版本

N/2 個人相關的問題
Alignment

輸入: 誰是MyGO!!!!!!的節奏吉他手?

輸出: 千早愛音

以剩下N/2 個人的問題進行測試

誰是MyGO!!!!!!的主唱?

LLM

高松燈

0% → ~80% 正確率



https://youtu.be/qycxA-xX_OY



【生成式AI】大模型 + 大資料 = 神奇結果？(2/3)：到底要多少資料才夠

現在 Pretrain 都用多大的資料？

LLaMA 3

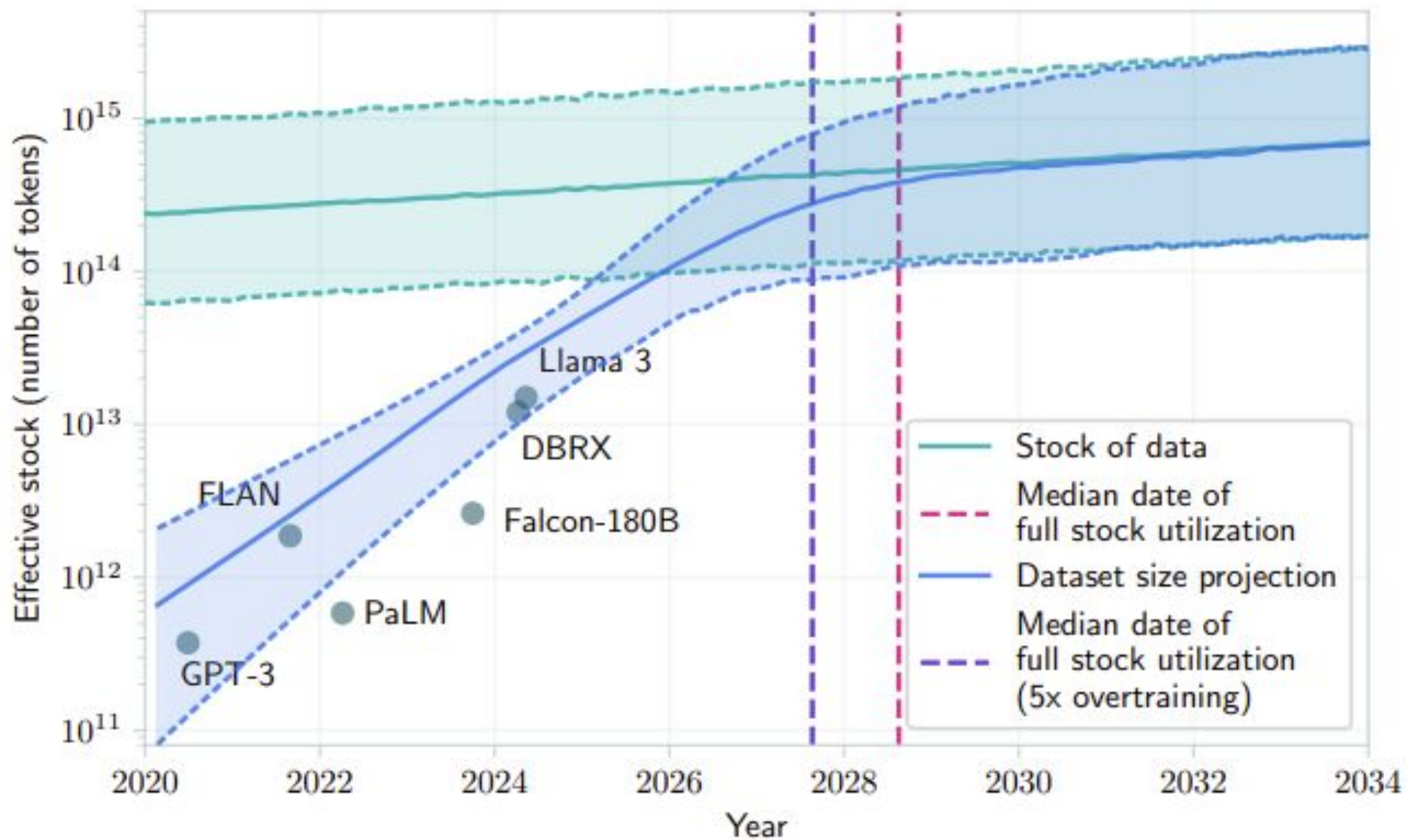
<https://arxiv.org/abs/2407.21783>

- **Data.** Compared to prior versions of Llama (Touvron et al., 2023a,b), we improved both the quantity and quality of the data we use for pre-training and post-training. These improvements include the development of more careful pre-processing and curation pipelines for pre-training data and the development of more rigorous quality assurance and filtering approaches for post-training data. We pre-train Llama 3 on a corpus of about 15T multilingual tokens, compared to 1.8T tokens for Llama 2.

DeepSeek-V3

<https://arxiv.org/abs/2412.19437>

We present DeepSeek-V3, a strong Mixture-of-Experts (MoE) language model with 671B total parameters with 37B activated for each token. To achieve efficient inference and cost-effective training, DeepSeek-V3 adopts Multi-head Latent Attention (MLA) and DeepSeekMoE architectures, which were thoroughly validated in DeepSeek-V2. Furthermore, DeepSeek-V3 pioneers an auxiliary-loss-free strategy for load balancing and sets a multi-token prediction training objective for stronger performance. We pre-train DeepSeek-V3 on 14.8 trillion diverse and high-quality tokens, followed by Supervised Fine-Tuning and Reinforcement Learning stages to fully harness its capabilities. Comprehensive evaluations reveal that DeepSeek-V3 outperforms other open-source models and achieves performance comparable to leading closed-source models. Despite its excellent performance, DeepSeek-V3 requires only 2.788M H800 GPU hours for its full training. In addition, its training process is remarkably stable. Throughout the entire training process, we did not experience any irrecoverable loss spikes or perform any rollbacks.



可以從哪裡取得大量資料



**15-trillion tokens,
44TB disk space**

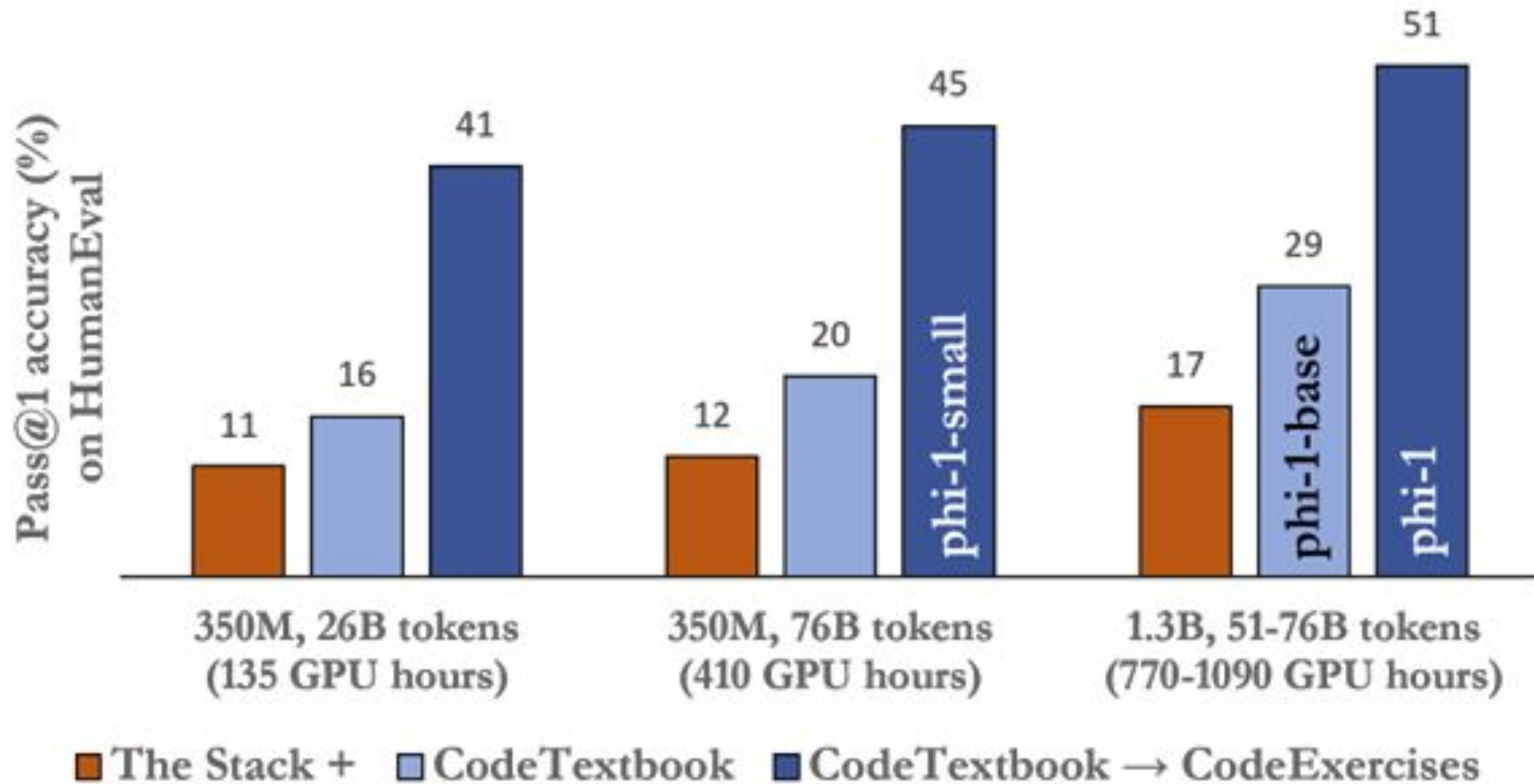
<https://arxiv.org/abs/2406.17557>

<https://huggingface.co/HuggingFaceFW>

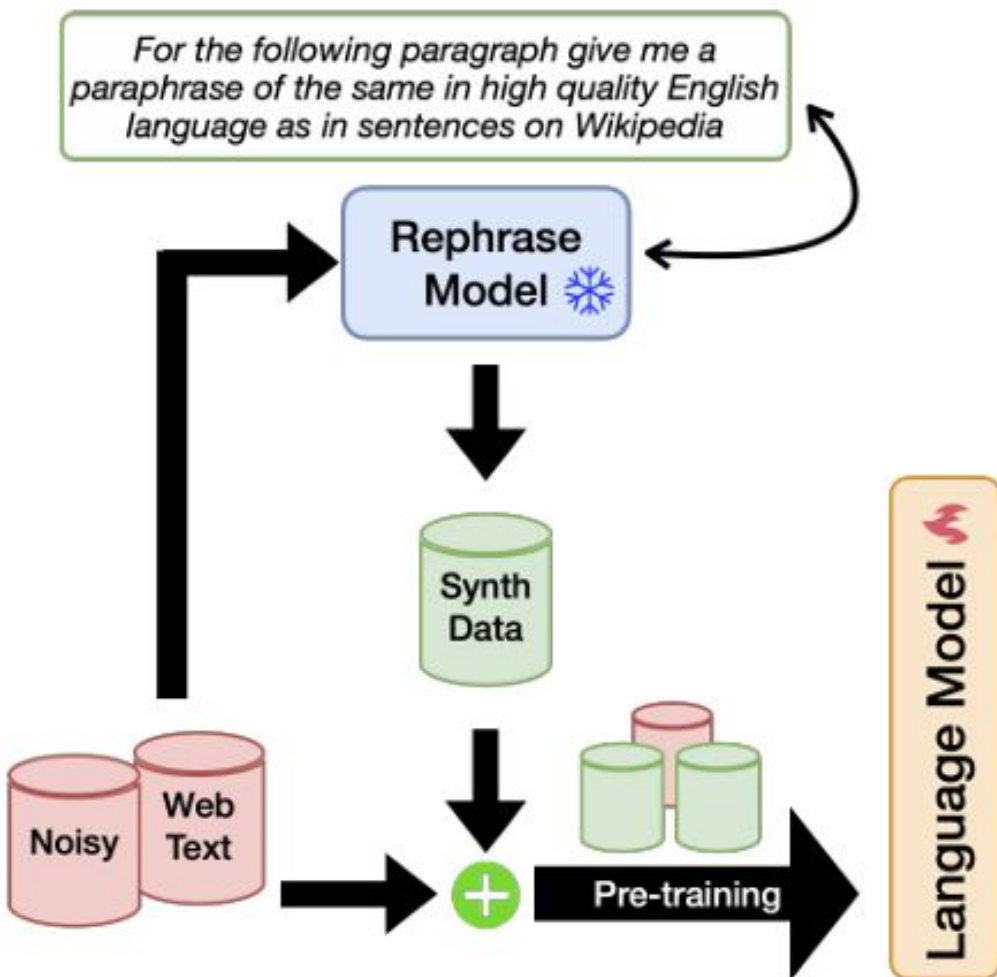
資料品質的重要性

Textbooks Are All You Need

<https://arxiv.org/abs/2306.11644>

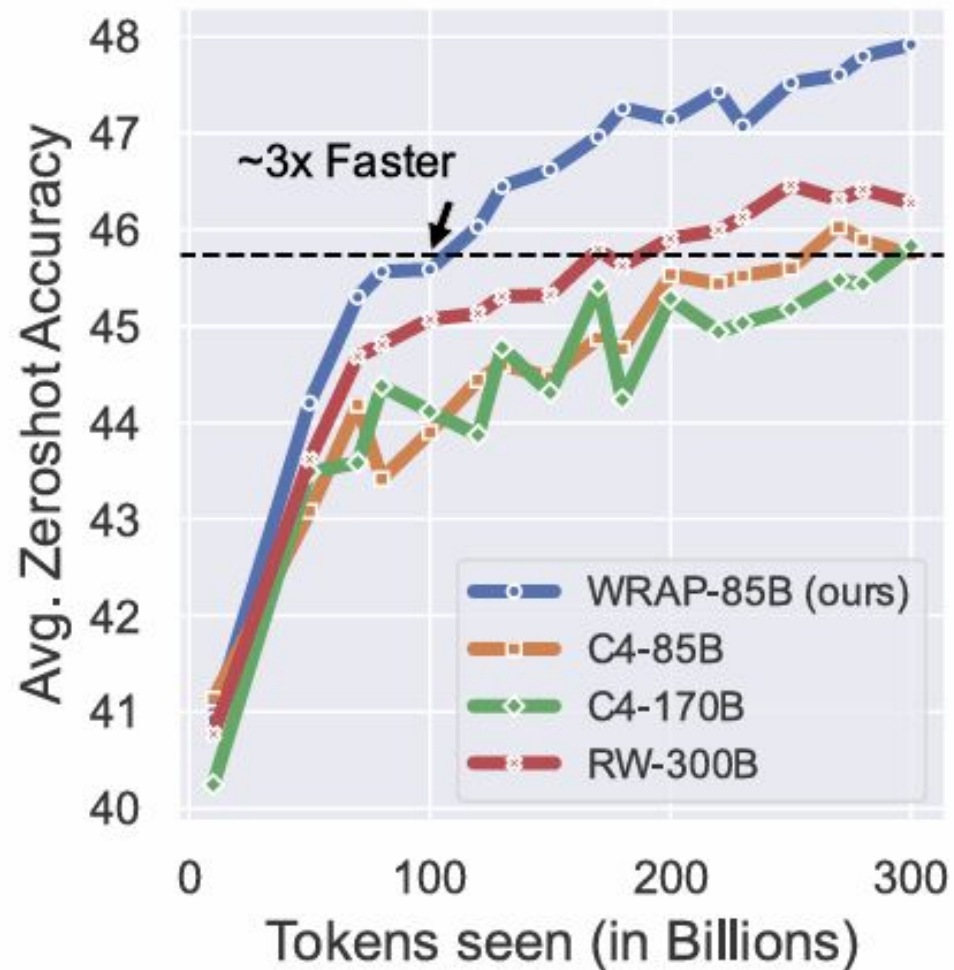


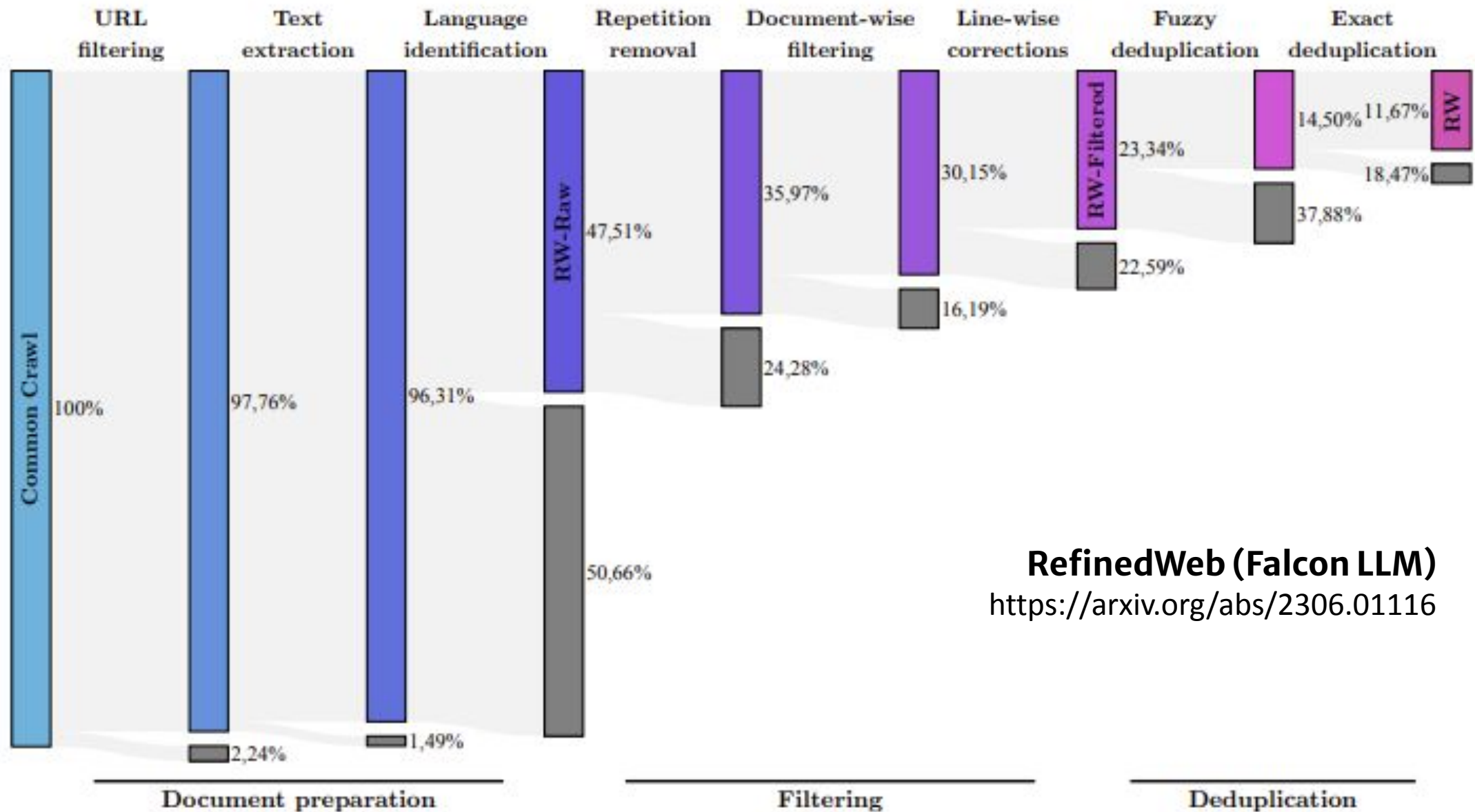
資料品質的重要性



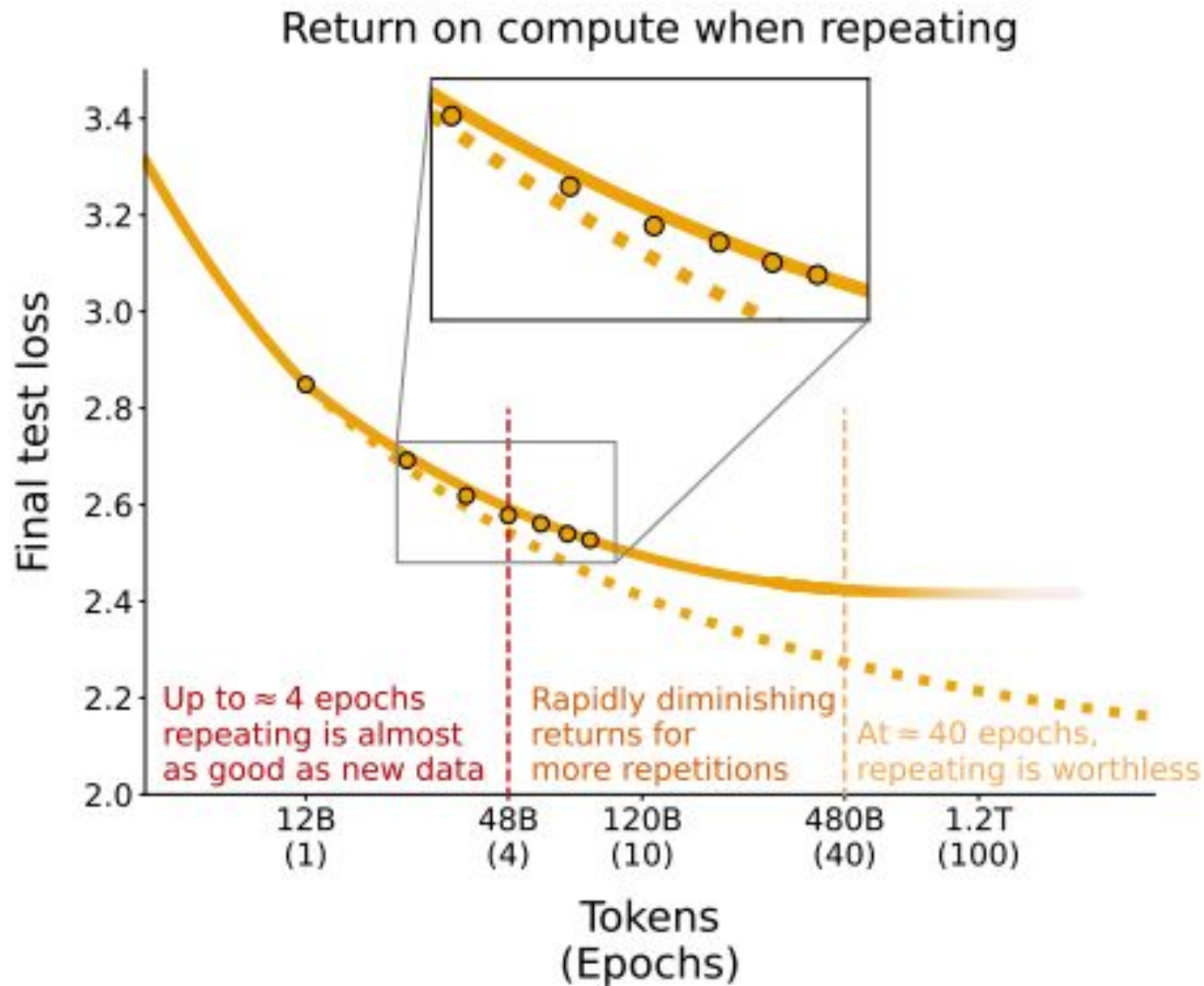
Rephrasing the Web

<https://arxiv.org/abs/2401.16380>





在有限算力、固定
模型下應該儘量
看更多不同的資料

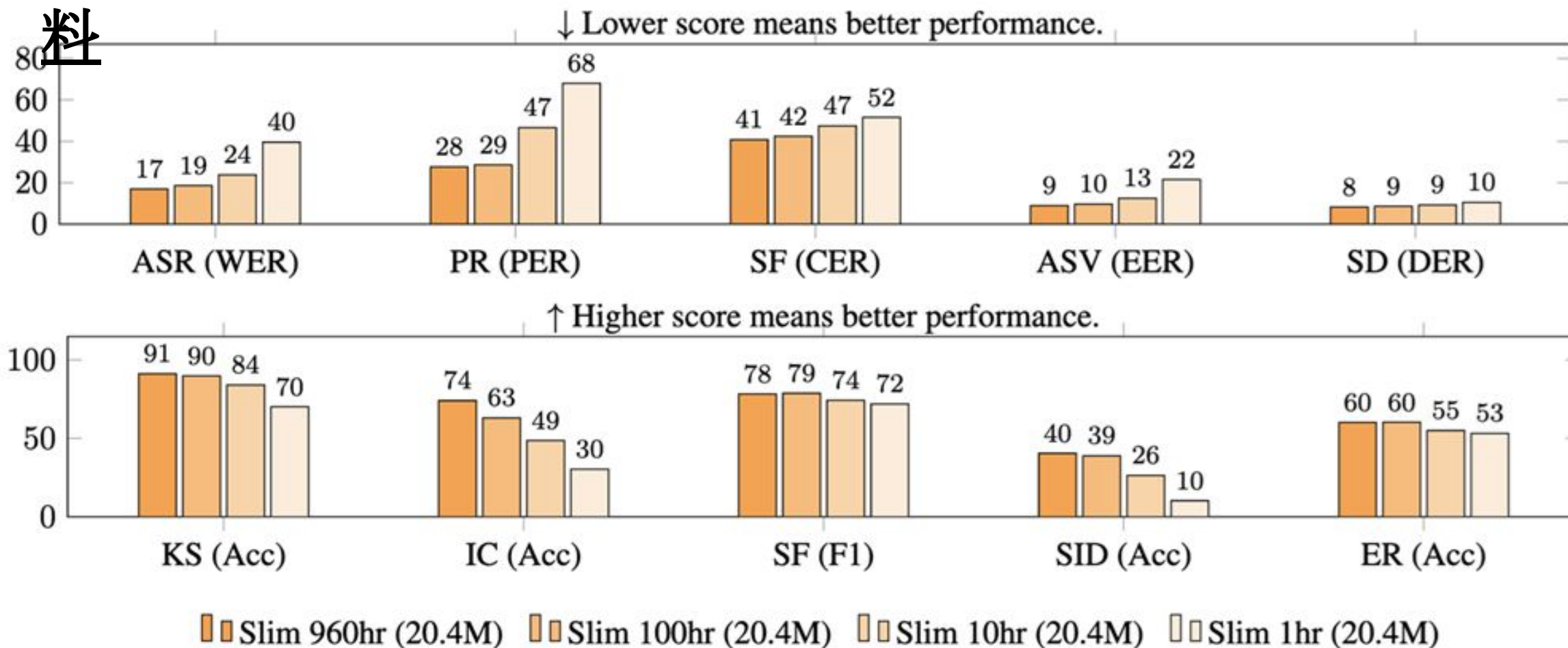


<https://arxiv.org/abs/2305.16264>

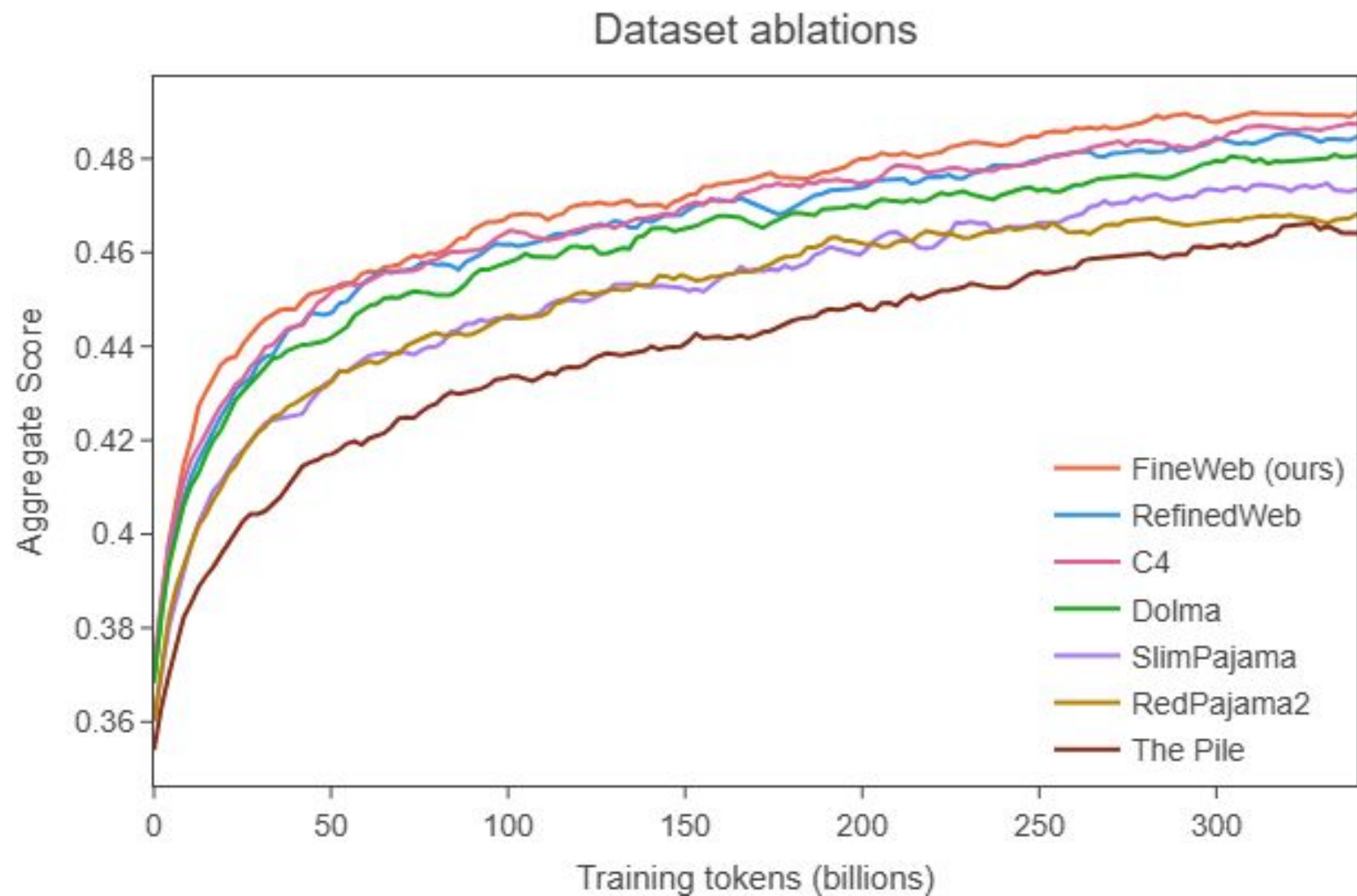
在有限算力、固定模型下應該儘量看更多不同的資料

Andy T. Liu

<https://arxiv.org/abs/2409.16295>



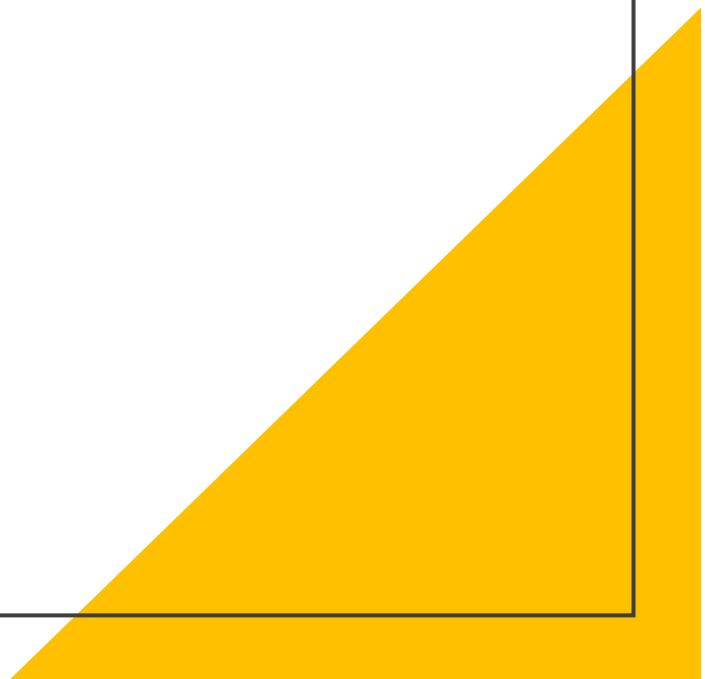
資料過濾



<https://arxiv.org/abs/2406.17557>

<https://huggingface.co/HuggingFaceFW>

Alignment 的極限



Query: How does actor critic improve over REINFORCE?

ChatGPT Response

Actor-critic algorithms are an extension of the REINFORCE algorithm that combine both policy-based and value-based methods. Some of the changes that actor-critic algorithms make over standard REINFORCE:

1. **Function Approximation:** Actor-critic algorithms typically use function approximation methods to represent the value and policy functions. This allows them to learn in high-dimensional state spaces.
 2. **Critic Network:** In actor-critic algorithms, there is an additional critic network that estimates the value function. The critic provides an estimate of the value of a state, which is then used to update the policy. This reduces the variance of the update and stabilizes learning.
 3. **Advantage Estimation:** Actor-critic algorithms use the advantage function to estimate the quality of an action taken by the policy. The advantage function is the difference between the return of an action and the estimated value of a state. This reduces the variance of the update and allows for more efficient learning.
- Overall, actor-critic algorithms provide a more efficient and stable way of learning in reinforcement learning tasks, particularly in high-dimensional state spaces.

Imitation Model Response

Actor-critic algorithms are a type of reinforcement learning algorithm that improves the REINFORCE algorithm by combining the actor (policy) and critic (value) components. The main changes that actor-critic algorithms make over standard REINFORCE are:

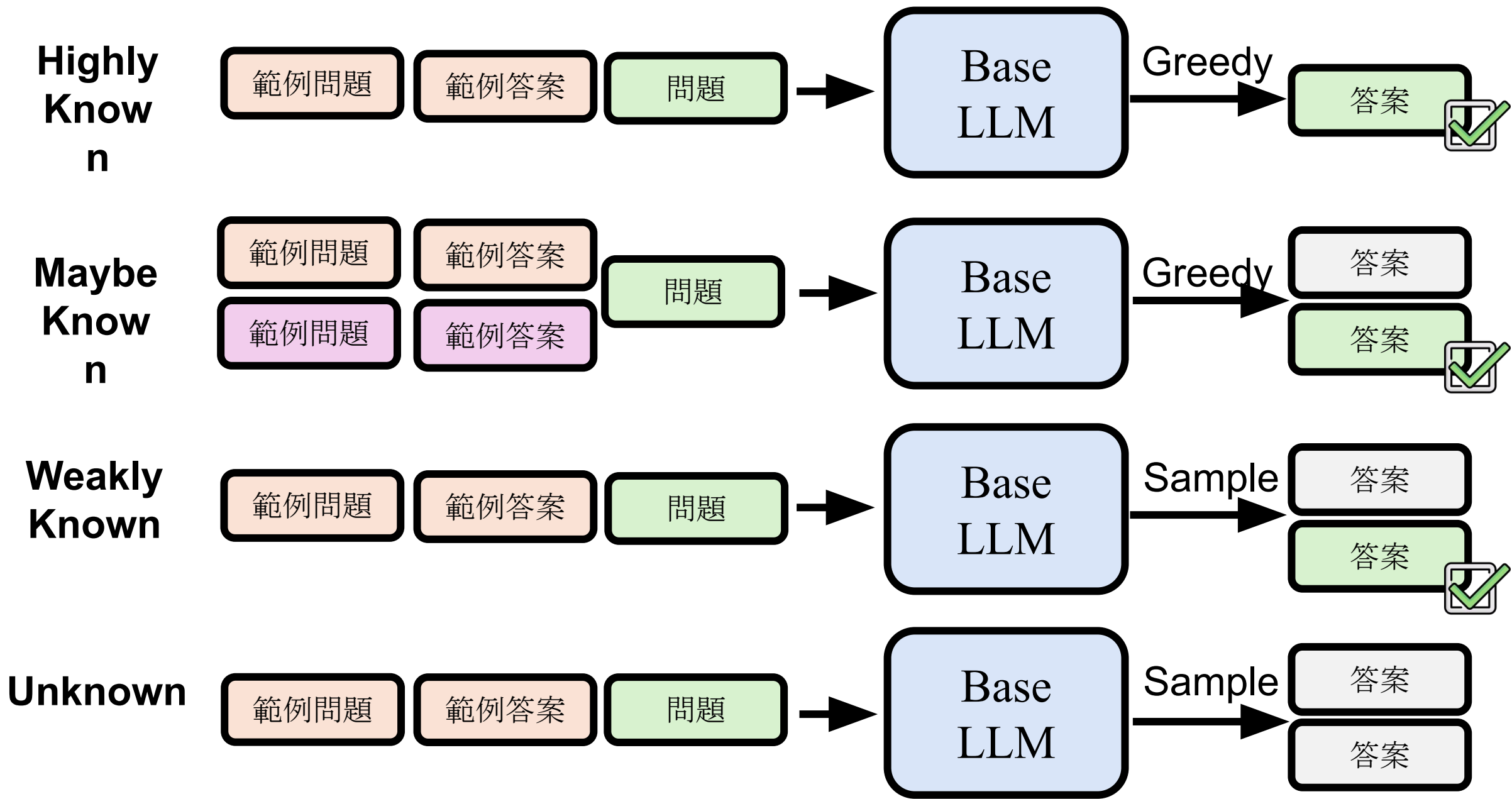
1. Actor-critic algorithms use a single critic to evaluate the value function of the policy, whereas standard REINFORCE uses a separate critic for each action.
 2. Actor-critic algorithms use a single actor to determine the actions to take, whereas standard REINFORCE uses a separate actor for each action.
 3. Actor-critic algorithms learn the policy and actor simultaneously, whereas standard REINFORCE learns the policy and actor separately.
 4. Actor-critic algorithms use an optimistic estimate of the value function to determine the actor's actions, whereas standard REINFORCE uses the critic's estimate.
- Overall, actor-critic algorithms combine the critic's and actor's estimates in order to learn both the policy and actor simultaneously, while standard REINFORCE learns them separately. This can lead to faster convergence and potentially better generalization, but the choice between the two algorithms depends on the specific problem and the desired behavior of the agent.

Finetune
LLaMA 1

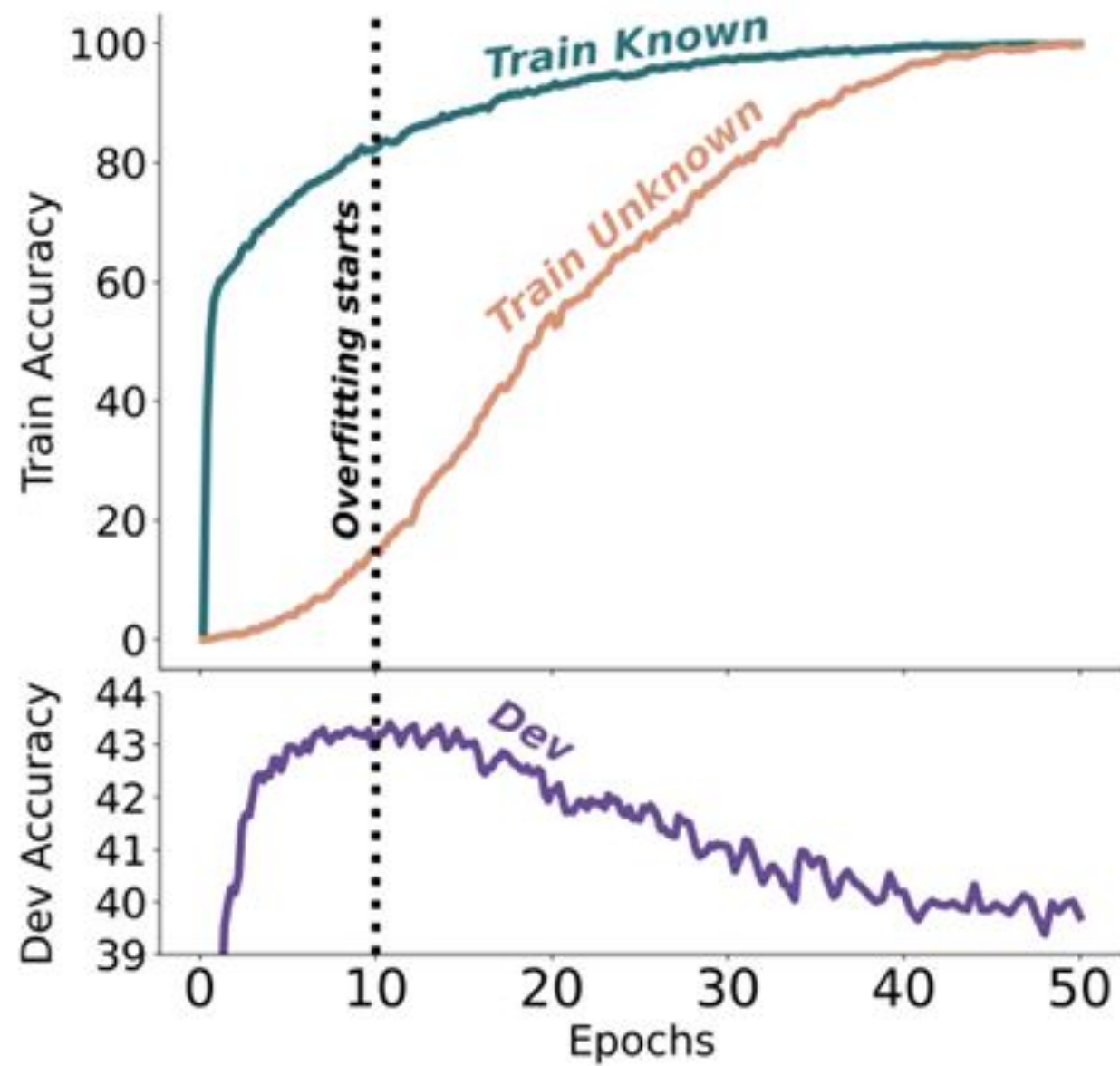


The False Promise of Imitating Proprietary LLMs

<https://arxiv.org/abs/2305.15717>



<https://arxiv.org/abs/2405.05904>



“MaybeKnown” 是最有幫助的

	EARLY_STOP					CONVERGENCE				
	Full	Hkn	Mkn	Wkn	Unk	Full	Hkn	Mkn	Wkn	Unk
$D_{\text{HighlyKnown}}$	40.5	98.7	60.1	9.0	0.6	40.0	98.4	58.8	8.5	0.7
$D_{\text{MaybeKnown}}$	43.6	98.4	69.9	12.1	1.0	43.2	97.5	68.2	12.9	1.3
$D_{\text{WeaklyKnown}}$	39.2	95.0	59.2	8.6	0.4	35.4	73.5	55.8	17.2	2.2
D_{Unknown}	37.5	95.6	52.9	6.5	0.6	25.8	55.8	36.6	12.2	3.2
D_{Natural}	43.5	98.0	67.6	14.1	1.8	41.8	95.5	61.7	14.8	2.5

<https://arxiv.org/abs/2405.05904>

Case 1

LLM本來就會的問題

LLM 自己的答案

Case 2

LLM不會的問題

正確答案

Case 3

LLM不會的問題

LLM 自己的答案

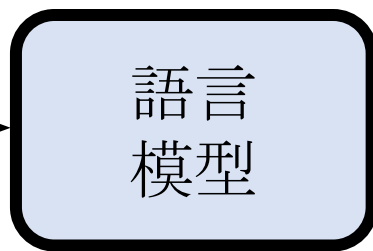
(錯誤答案)

Eval	Medicine			History			Engineering			Jurisprudence		
	HAR	INC	SELF	HAR	INC	SELF	HAR	INC	SELF	HAR	INC	SELF
LLaMA-2-7B												
HOMO	40.22	28.45	37.00	38.80	29.60	33.60	48.40	32.40	32.80	37.60	34.00	33.20
ID	39.82	37.26	41.46	54.30	31.08	46.02	42.07	31.03	26.21	38.86	35.70	36.34
OOD	39.97	36.75	40.94	39.64	30.69	37.22	40.38	28.26	29.17	38.49	34.56	34.88
LLaMA-2-13B												
HOMO	40.83	36.05	34.41	48.40	32.40	43.60	58.00	37.20	55.20	44.00	32.40	37.60
ID	55.43	35.06	52.13	68.28	46.13	64.09	45.52	29.66	40.00	54.77	38.55	52.77
OOD	54.21	35.77	50.98	51.30	37.98	49.06	52.15	35.94	51.12	50.83	39.26	48.27
LLaMA-2-70B												
HOMO	47.95	42.54	46.03	59.20	42.00	51.60	62.40	55.20	57.60	55.20	47.60	51.60
ID	65.37	61.40	63.11	82.37	71.29	81.29	55.17	39.31	54.48	67.69	62.21	67.52
OOD	65.34	60.35	63.93	63.63	57.94	63.54	65.62	59.21	64.75	61.90	57.03	61.45
Mistral-7B												
HOMO	49.80	34.68	35.02	46.80	33.20	40.80	59.60	48.40	55.20	48.00	38.80	43.60
ID	58.17	41.77	51.83	67.74	29.35	50.11	44.83	31.03	42.07	55.21	41.43	49.38
OOD	54.48	40.47	47.81	53.07	32.98	45.07	50.49	41.89	44.51	52.42	40.93	48.88

RL 是 Alignment 的好方法

這不是人類強制給予的答案，
這是語言模型自己生成的答案

台灣最高的山是那座？



玉山



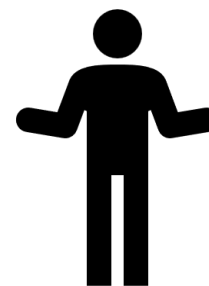
誰來告訴我呀



提高機率



減低機率



Pretrain 的後遺症？

GPT 4.5 都可以答對

Shift ciphers

Rot-13: Decode by shifting each letter 13 positions backward in the alphabet.

Input: Ohg guvf gvzr, gurer znl nyfb or nabgure ernfba.

Correct: But this time, there may also be another reason.

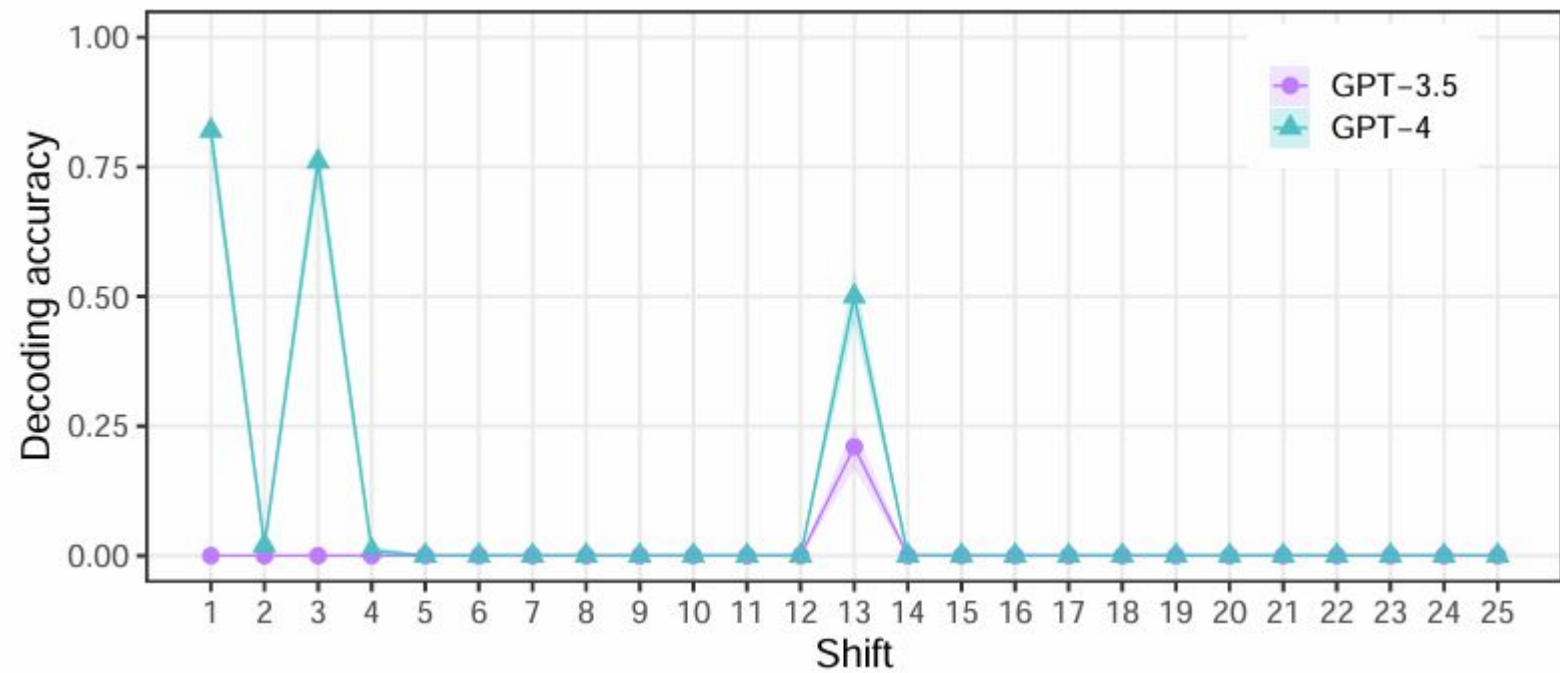
✓ GPT-4: But this time, there may also be another reason.

Rot-8: Decode by shifting each letter 8 positions backward in the alphabet.

Input: Jcb bpqa bqum, bpmzm uig itaw jm ivwbpmz zmiawv.

Correct: But this time, there may also be another reason.

✗ GPT-4: Say what you, think and then be silent.



Embers of Autoregression

<https://arxiv.org/abs/2309.13638>



全部

影片

地圖

提示： 限制搜尋



Wikipedia
<https://en.wikipedia.org>

ROT13

ROT13 is a simple
alphabet.



dCode
<https://www.dcode.fr>

shift cipher |



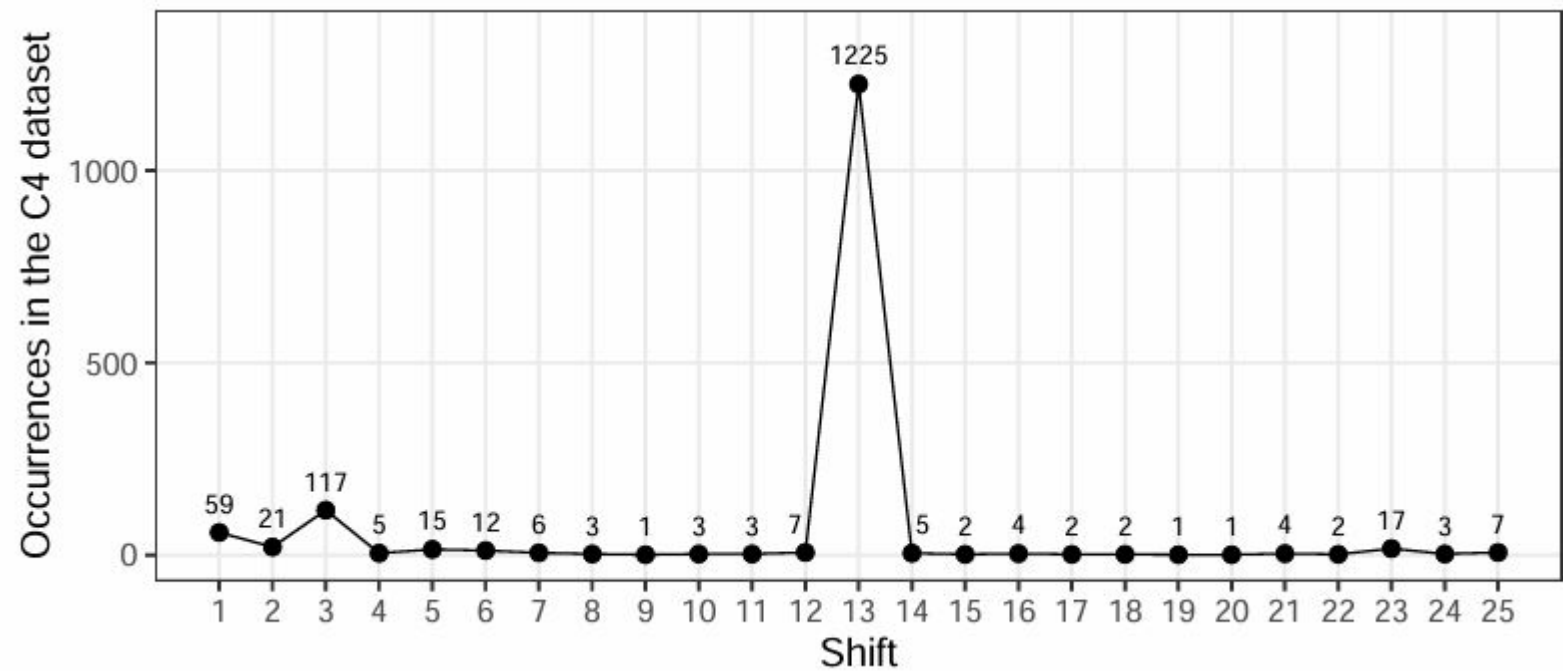
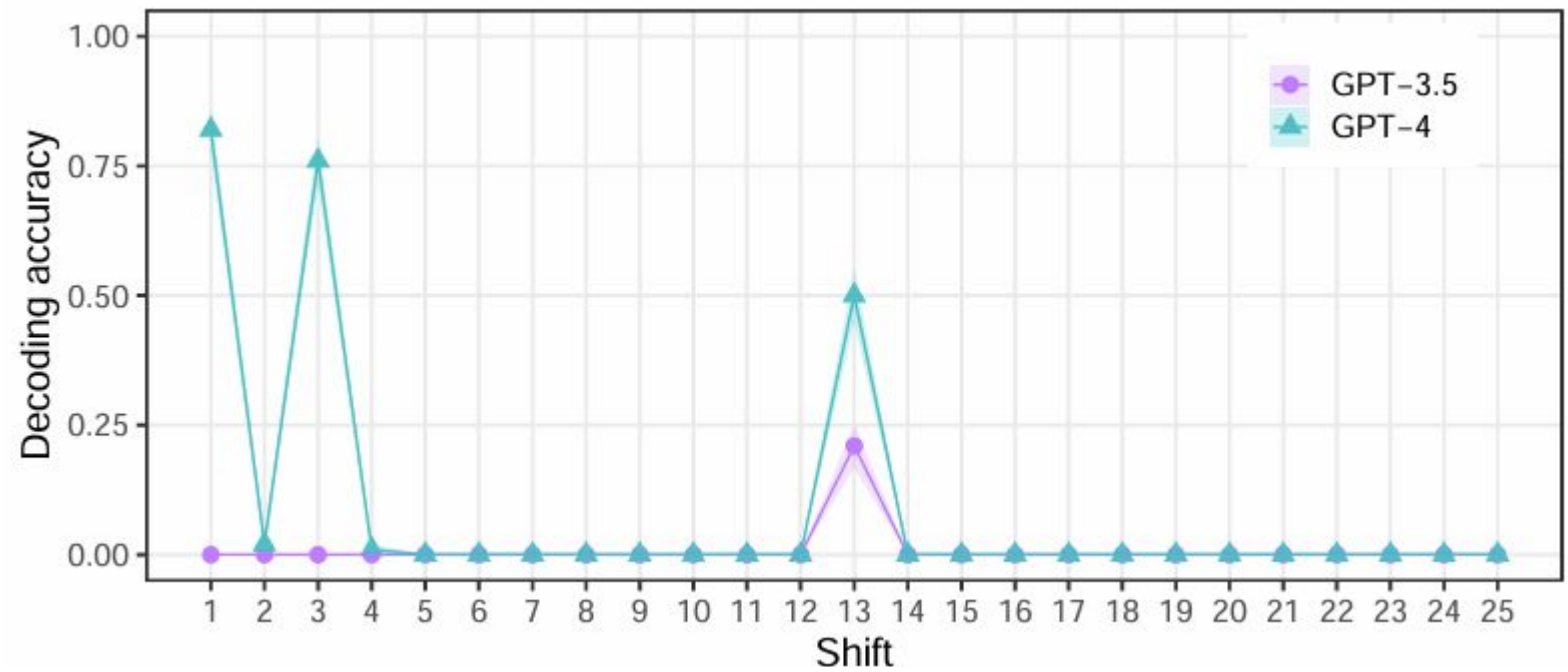
- shift cipher decoder
- shift cipher 13
- shift cipher in cryptography
- shift cipher solver
- shift cipher calculator
- shift cipher example
- shift cipher python
- shift cipher wheel
- shift cipher definition
- shift cipher formula

回報不適當的預測查詢字串



ROT-13 Cipher - ROT13 - Online Text Decoder, Encoder, ...

Rot-13 (short for Rotation 13) is the name given to a mono-alphabetical substitution cipher which has the property of being reversible and very simple.

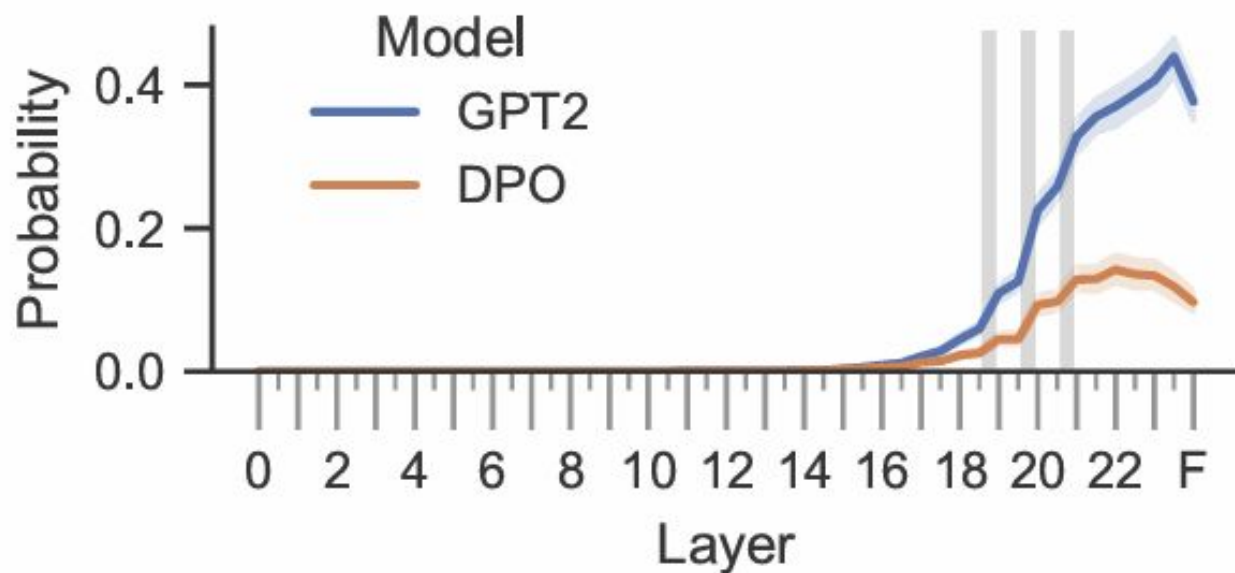


Embers of Autoregression

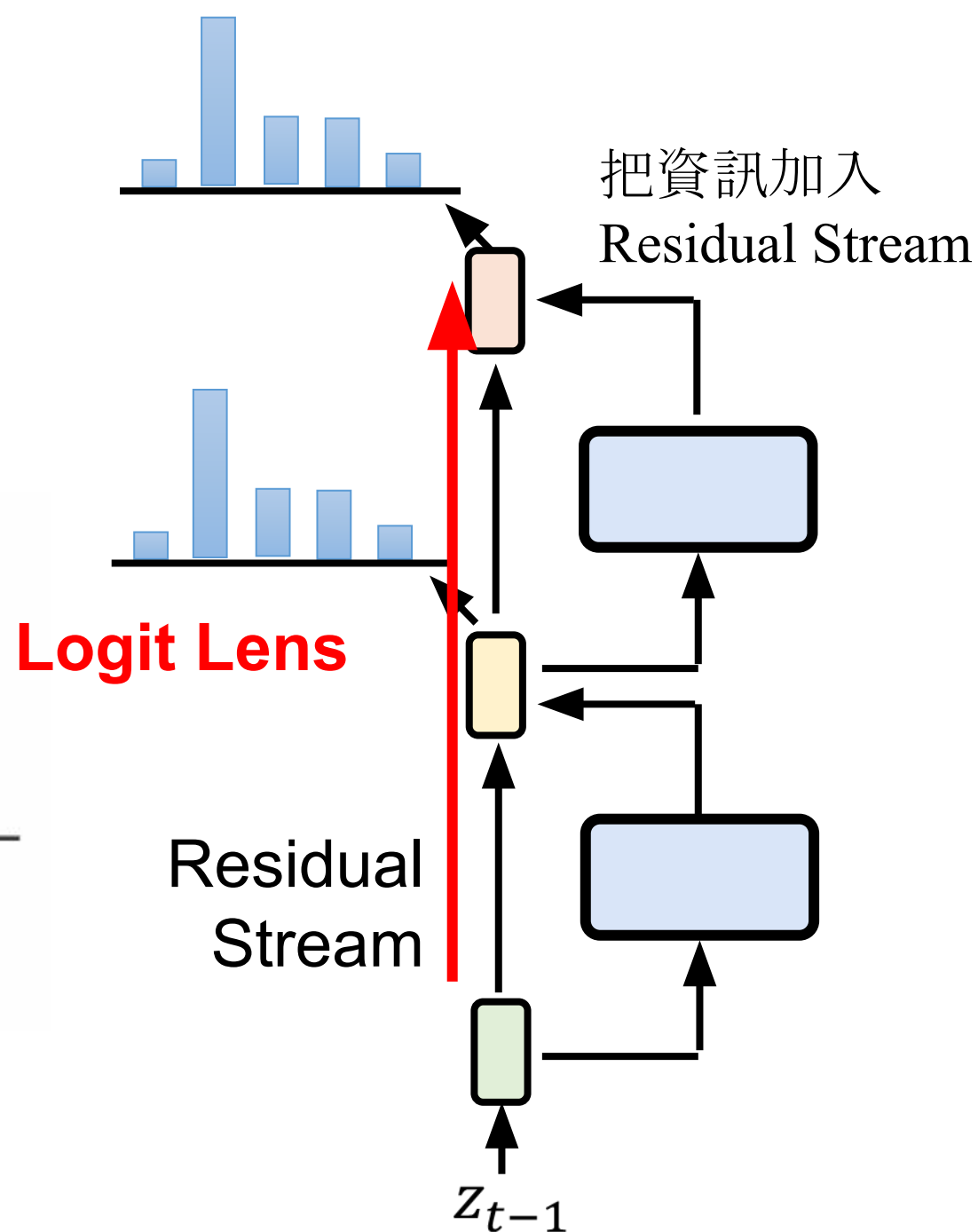
<https://arxiv.org/abs/2309.13638>

Pretrain 時看到不該看的東西後，難以真正清除

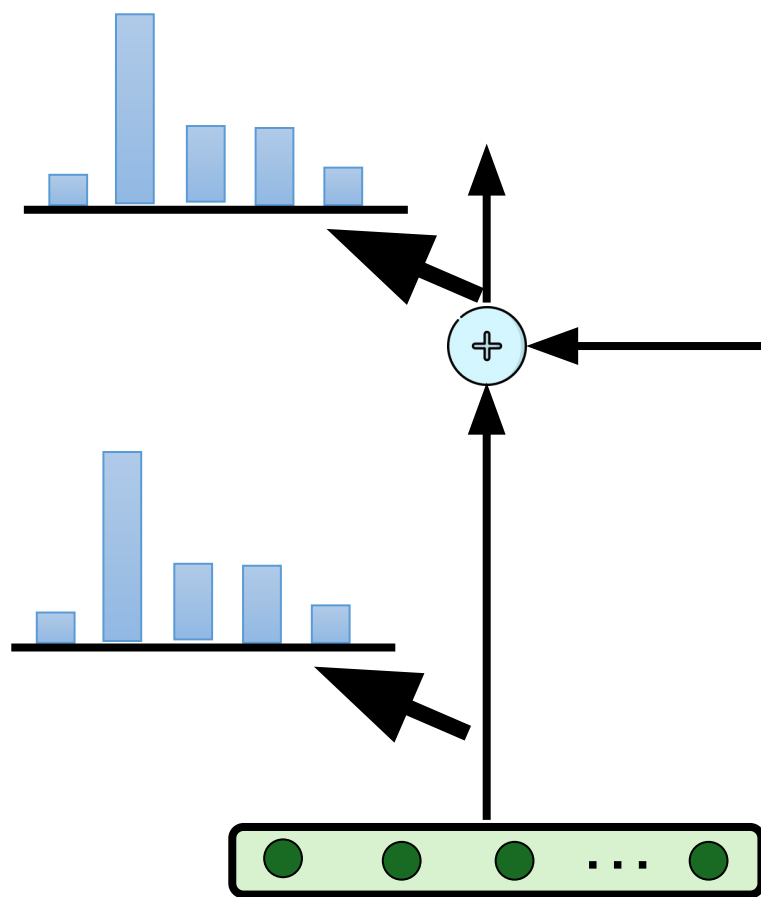
Sh*t 的機率



<https://arxiv.org/abs/2401.01967>

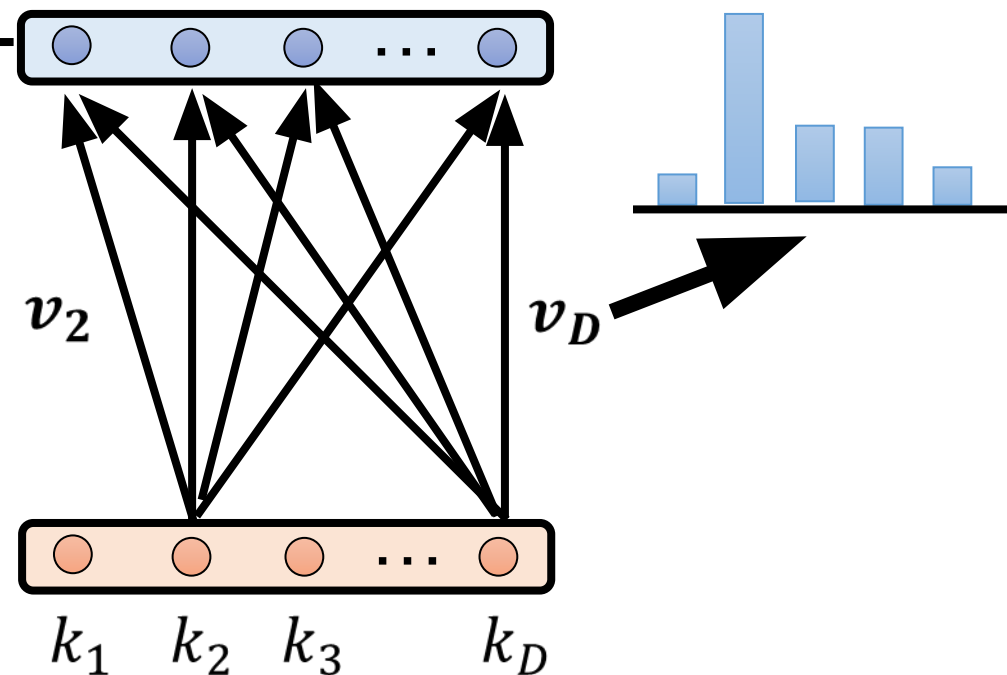


Pretrain 時看到不該看的東西後，難以真正清除



MLP.v₇₇₀¹⁹
MLP.v₇₇₁¹²
MLP.v₂₆₆₉¹⁸
MLP.v₆₆₈¹³
MLP.v₂₅₅¹⁶
MLP.v₈₈₂¹²
MLP.v₁₄₃₈¹⁹

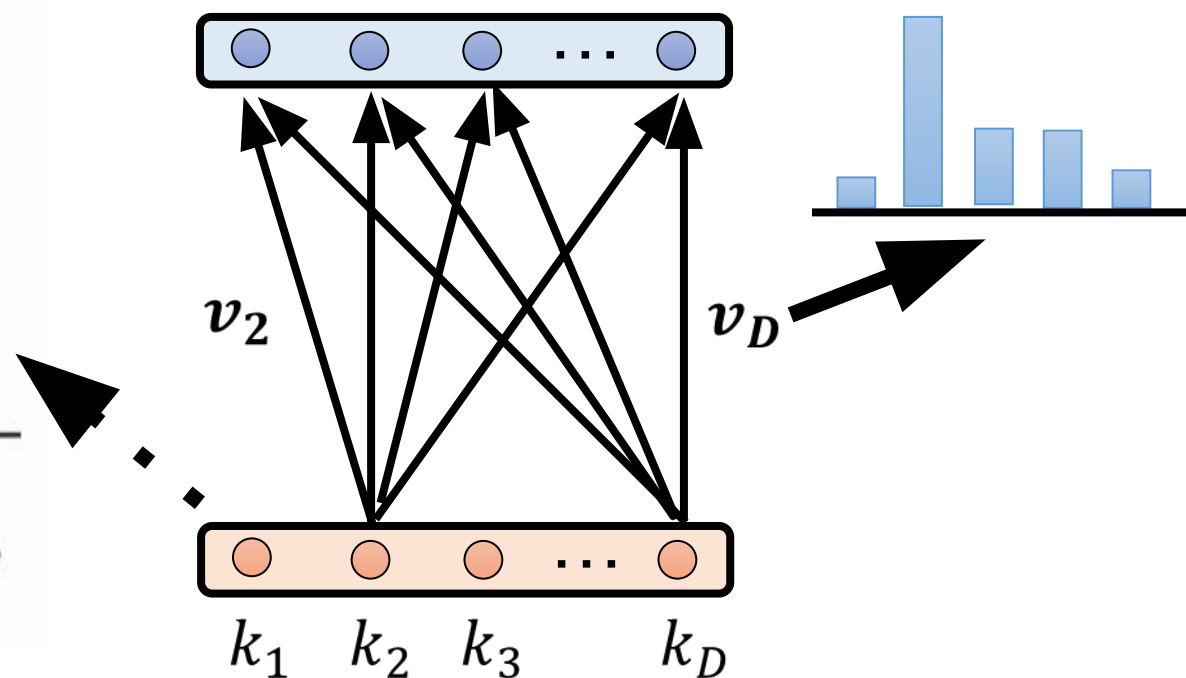
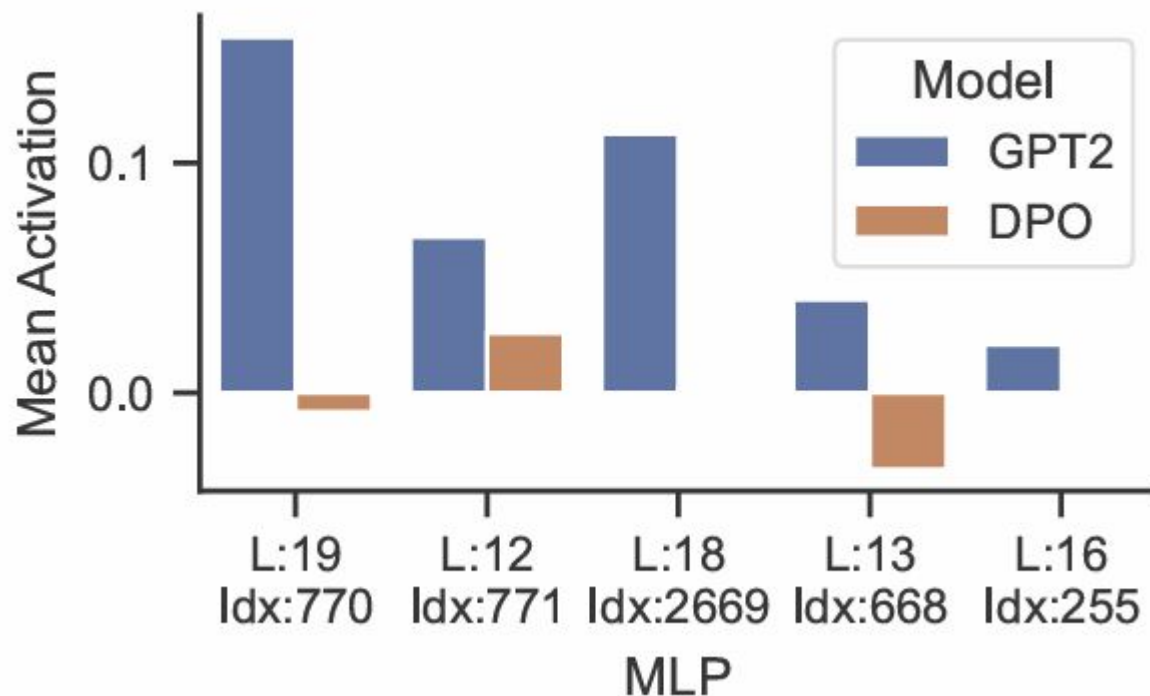
sh*t, a**, cr*p, f*ck, c*nt, garbage, trash
delusional, hypocritical, arrogant, nonsense
degener, whining, idiots, stupid, smug
losers, filthy, disgr, gad, feces, apes, thous
disgrace, shameful, coward, unacceptable
f*ck, sh*t, piss, hilar, stupidity, poop
c*m, c*ck, orgasm, missionary, anal



Pretrain 時看到不該看的東西後，難以真正清除

MLP.v₇₇₀¹⁹
MLP.v₇₇₁¹²
MLP.v₂₆₆₉¹⁸
MLP.v₆₆₈¹³
MLP.v₂₅₅¹⁶
MLP.v₈₈₂¹²
MLP.v₁₄₃₈¹⁹

sh*t, a**, cr*p, f*ck, c*nt, garbage, trash
delusional, hypocritical, arrogant, nonsense
degener, whining, idiots, stupid, smug
losers, filthy, disgr, gad, feces, apes, thous
disgrace, shameful, coward, unacceptable
f*ck, sh*t, piss, hilar, stupidity, poop
c*m, c*ck, orgasm, missionary, anal



Alignment



Pretrain

idea from: <https://x.com/anthrupad/status/1622349563922362368/photo/1>

結語

- Pretrain-Alignment 很強大
 - LLM 在 Pretrain 已經很強, Alignment 只是畫龍點睛
 - Pretrain 階段看過大量各式各樣的資料是關鍵
- Pretrain-Alignment 有極限
 - 在 Alignment 階段往往 LLM 只是強化原來已經知道的事情, 難以學習新技能

下集預告: 如何有效微調模型